

Lecture5b

September 29, 2025

1 CS5489 - Machine Learning

2 Lecture 5b - Neural Networks

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```
[1]: # setup
%matplotlib inline
import matplotlib_inline # setup output image format
matplotlib_inline.backend_inline.set_matplotlib_formats('retina')
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 100 # display larger images
import matplotlib
from numpy import *
from sklearn import *
from scipy import stats

rbow = plt.get_cmap('rainbow')
```

3 Outline

- History
- Perceptron
- Multi-class logistic regression
- **Multi-layer perceptron (MLP)**

4 Extracting features

- The multi-class logistic regression model assumes the inputs are feature vectors.
 - $\mathbf{g} = \mathbf{W}^T \mathbf{x}$
 - $\mathbf{f} = \mathbf{s}(\mathbf{g})$
- What if we also want to learn the feature vectors?
 - Replace $\mathbf{x}$ with a feature extractor.
 - For simplicity, we can reuse the same “unit” as the classifier to compute the “features”.
- Replace $\mathbf{x}$ with feature extractor $\mathbf{z} = \sigma(\mathbf{A}^T \mathbf{x})$
 - $\mathbf{z}$ is the extracted feature vector (also called *hidden* nodes)

- * $z_i = \sigma(\mathbf{a}_i^T \mathbf{x})$ is a hidden node.
- $\sigma()$ is the sigmoid function (also called *activation* function.)
- $\mathbf{A}$ are the parameters of the feature extractor.
- New model
 - $\mathbf{z} = \sigma(\mathbf{A}^T \mathbf{x})$
 - $\mathbf{g} = \mathbf{W}^T \mathbf{z}$
 - $\mathbf{f} = \mathbf{s}(\mathbf{g})$
- Interpretation
 - Each hidden node z_i is a “classifier” looking for pattern based on $\mathbf{a}_i$.
 - The output is looking for patterns in the vector $\mathbf{x}$
 - the effect is a non-linear classifier in the input space of $\mathbf{x}$.
 - we can apply this recursively to create features of features ...

5 Multi-layer Perceptron

- Add hidden layers between the inputs and outputs
 - each hidden node is a Perceptron (with its own set of weights)
 - * its inputs are the outputs from previous layer
 - * extracts a feature pattern from the previous layer
 - can model more complex functions
- Formally, for one layer:
 - $\mathbf{h} = f(\mathbf{W}^T \mathbf{x})$
 - * Weight matrix $\mathbf{W}$ - one column for each node
 - * Input $\mathbf{x}$ - from previous layer
 - * Output $\mathbf{h}$ - to next layer
 - * $f(a)$ is the activation function - applied to each dimension to get output
- Also called *fully-connected layers* or *dense layers*

6 Activation functions

- There are different types of activation functions:
 - *Sigmoid* - output $[0,1]$
 - *Tanh* - output $[-1,1]$
 - *Rectifier Linear Unit (ReLU)* - output $[0,\infty]$

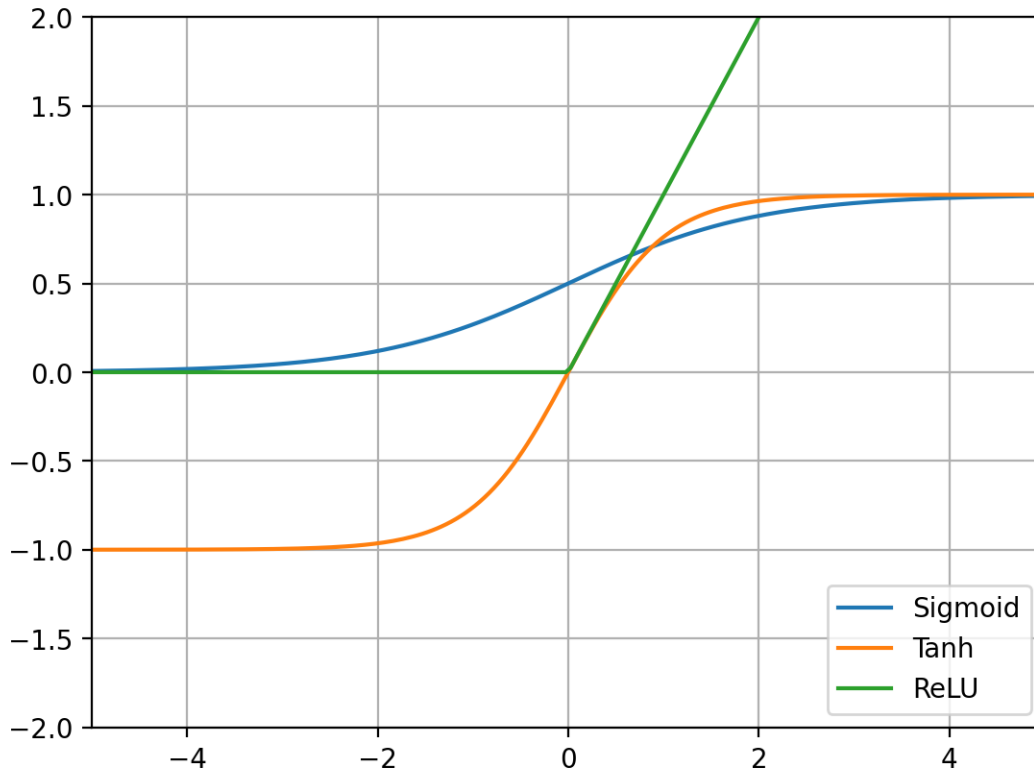
```
[2]: x=linspace(-5,5,200)
fs = {'Sigmoid': lambda x: 1/(1+exp(-x)),
      'Tanh':    lambda x: tanh(x),
      'ReLU':    lambda x: maximum(0,x),
      }

actfig = plt.figure()
for name,f in fs.items():
    plt.plot(x, f(x), label=name)
plt.legend(loc="lower right")
```

```
plt.axis([-5,5,-2,2])
plt.grid(True)
plt.close()
```

```
[3]: actfig
```

```
[3]:
```



- Activation functions specifically for output nodes:
 - *Linear* - output for regression
 - *Softmax* - output for classification (same as multi-class logistic regression)
- Each layer can use a different activation function.

7 Which activation function is best?

- In the early days, only the Sigmoid and Tanh activation functions were used.
 - these were notoriously hard to train with.
 - * gradient decays to zero on both ends
- Recently, ReLU has become very popular.
 - easier to train with - gradient is either 0 or 1.
 - faster - don't need to calculate exponential
 - sparse representation - most nodes will output zero.

8 Training an MLP

- Assume a general case:
 - linear transform: $\mathbf{g}_1 = \mathbf{A}^T \mathbf{x}$
 - activation: $\mathbf{z} = h_1(\mathbf{g}_1)$
 - linear transform: $\mathbf{g}_2 = \mathbf{W}^T \mathbf{z}$
 - activation: $\mathbf{f} = h_2(\mathbf{g}_2)$
 - loss function: $L(\mathbf{y}, \mathbf{f})$
- In our example, h_1 is a sigmoid, h_2 is a softmax.
- Similar to multi-class logistic regression
 - minimize the loss
 - * $(\mathbf{A}^*, \mathbf{W}^*) = \operatorname{argmin}_{\mathbf{A}, \mathbf{W}} L(\mathbf{y}, \mathbf{f}(\mathbf{x}))$
 - use gradient descent as before
 - * need to compute $\frac{dL}{d\mathbf{A}}$ and $\frac{dL}{d\mathbf{W}}$
 - * we have done most of the work already...
- Computation graph
- Chain rule:
 - 1) $\frac{dL}{d\mathbf{f}}$
 - 2) $\frac{dL}{d\mathbf{g}_2} = \frac{d\mathbf{f}^T}{d\mathbf{g}_2} \frac{dL}{d\mathbf{f}} \Rightarrow$ 3) $\frac{dL}{d\mathbf{w}_j} = \frac{d\mathbf{g}_2^T}{d\mathbf{w}_j} \frac{dL}{d\mathbf{g}_2}$
 - 4) $\frac{dL}{d\mathbf{z}} = \frac{d\mathbf{g}_2^T}{d\mathbf{z}} \frac{dL}{d\mathbf{g}_2}$
 - 5) $\frac{dL}{d\mathbf{g}_1} = \frac{d\mathbf{z}^T}{d\mathbf{g}_1} \frac{dL}{d\mathbf{z}} \Rightarrow$ 6) $\frac{dL}{d\mathbf{a}_j} = \frac{d\mathbf{g}_1^T}{d\mathbf{a}_j} \frac{dL}{d\mathbf{g}_1}$
- Recursively use the gradient of descendant layers
 - propagate gradient backwards
 - backwards propagation
 - back propagation
 - backpropagation
 - backprop
 - BP

9 Backpropagation (backward propagation)

- Defines a set of recursive relationships
 - 1) calculate the output of each node from first to last layer
 - 2) calculate the gradient of each node from last to first layer
- **NOTE:** the gradients multiply in each layer!
 - if two gradients are small (<1), their product will be even smaller. This is the “vanishing gradient” problem.

10 Backpropagation (general form)

- Given a computation graph
 - 1) apply input $\mathbf{x}$ and forward propagate to compute all the nodes' values and the loss.
 - 2) Go backwards from end, at node $\mathbf{h}$:
 - compute gradients from child nodes: $\frac{dL}{d\mathbf{h}} = \sum_{g \in \text{child}(\mathbf{h})} \frac{d\mathbf{g}^T}{d\mathbf{h}} \frac{dL}{d\mathbf{g}}$

- compute gradient of parameters $\mathbf{w}_h$ of h : $\frac{dL}{d\mathbf{w}_h} = \frac{d\mathbf{h}^T}{d\mathbf{w}_h} \frac{dL}{d\mathbf{h}}$
- This is the “symbol-to-number” differentiation (e.g., Caffe, Torch).
- We can also make the backprop operations as part of the computation graph.
 - called “symbol-to-symbol” differentiation (e.g., Tensorflow, Theano)

11 Stochastic Gradient Descent (SGD)

- The datasets needed to train NN are typically very large
- Use SGD so that only a small portion of the dataset is needed at a time
 - Each small portion is called a *mini-batch*
 - Use a *momentum* term, which averages the current gradient with those from previous mini-batches.
 - One complete pass through the data is called an *epoch*.

12 Monitoring training with SGD

- Separate the training set into training and validation
 - use the training set to run backpropagation
 - test the NN on the validation set for diagnostics
 - * check for convergence - adjust learning rate if necessary
 - * check for diverging loss - adjust learning rate
 - * stopping criteria - stop when no change in the validation error.
 - * decay learning rate after each epoch.

13 Load NN software

- We will use *pytorch*
 - compatible with scikit-learn
 - pytorch is an easy-to-use and widely-utilized framework

```
[4]: import random
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
```

```
[5]: import sys
print("Python:", sys.version, "Torch", torch.__version__)
from typing import List
```

```
Python: 3.13.7 | packaged by Anaconda, Inc. | (main, Sep  9 2025, 19:54:17)
[Clang 17.0.6 ] Torch 2.8.0
```

```
[6]: # initialize random seed
torch.manual_seed(4487)
np.random.seed(4487)
```

```
random.seed(4487)
```

```
[7]: # type three - moons
X0,Y = datasets.make_moons(n_samples=100, noise=0.1, random_state=4487)
mycmap = matplotlib.colors.LinearSegmentedColormap.from_list('mycmap',
    ↪["#FF0000", "#FFFFFF", "#00FF00"])

scaler = preprocessing.MinMaxScaler(feature_range=(-1,1))
X = scaler.fit_transform(X0)

axbox = [-1.2, 1.2, -1.2, 1.2]
```

```
[8]: def plot_nn(model, axbox, X, Y, mode="prob"):
    # assumes a sigmoid output node
    # grid points
    xr = [ linspace(axbox[0], axbox[1], 200),
           linspace(axbox[2], axbox[3], 200) ]

    # make a grid for calculating the posterior,
    # then form into a big [N,2] matrix
    xgrid0, xgrid1 = meshgrid(xr[0], xr[1])
    allpts = torch.as_tensor(c_[xgrid0.ravel(), xgrid1.ravel()], dtype=torch.
    ↪float32)

    # print(xgrid0.shape, xgrid1.shape, allpts.shape)

    # calculate the decision function
    if mode == "prob":
        #scoreall = model.predict_proba(allpts) # tf1
        # score = scoreall[:,1].reshape(xgrid0.shape)
        scoreall = model(allpts).softmax(-1)
        score = scoreall[:,1].view(xgrid0.shape).detach().numpy()
        maxscore = 1.0
        minscore = 0.0
        al = 0.3

    plt.imshow(score, origin='lower', extent=axbox, alpha=al, cmap=mycmap,
               vmin=minscore, vmax=maxscore, aspect='auto')

    plt.contour(xr[0], xr[1], score, levels=[0.5], linestyles='solid',
    ↪colors='k',
               linewidths=2)

    plt.scatter(X[:,0], X[:,1], c=Y, cmap=mycmap, edgecolors='k')
    plt.axis(axbox); plt.grid(True)
```

```
[9]: def plot_history(history):
    plt.plot(history['train_loss'], label="training loss ({:.6f})".
    ↪format(history['train_loss'][-1]))
    plt.plot(history['val_loss'], label="validation loss ({:.6f})".
    ↪format(history['val_loss'][-1]))
    plt.grid(True)
    plt.legend(loc="best", fontsize=9)
    plt.axis([0, len(history['train_loss']), 0, 0.7])
    plt.xlabel('iteration')
    plt.ylabel('loss')
```

```
[10]: # split training and validation
val_num = int(len(X) * 0.1) # 10% as validation
arr_permute = np.random.permutation(len(X))
X, Y = X[arr_permute], Y[arr_permute]
trainX, trainY = X[:-val_num].astype(np.float32), Y[:-val_num]
valX, valY = X[-val_num:].astype(np.float32), Y[-val_num:]
```

Let's define a NN module and a training loop with pytorch!

```
[11]: # define the network
class MyModel(nn.Module):
    def __init__(self, input_size: int, output_size: int, hidden_sizes:
    ↪List[int]):
        super().__init__()
        self.input_size, self.output_size, self.hidden_sizes = input_size,
    ↪output_size, hidden_sizes
        layers = []
        # for the hidden layers
        prev_size = input_size
        for curr_size in hidden_sizes:
            layers.append(nn.Linear(prev_size, curr_size)) # [bs, hidden_size]
            layers.append(nn.ReLU()) # we use relu activation
            prev_size = curr_size
        # for the output layer
        layers.append(nn.Linear(prev_size, output_size)) # [bs, output_size]
        # layers.append(nn.Softmax(dim=-1)) # we perform softmax at the
    ↪outside, not here!
        # add to the Module itself
        self.network = nn.Sequential(*layers)

    def forward(self, x):
        t_input = x.view(-1, self.input_size) # [bs, input_size]
        t_output = self.network(t_input) # [bs, output_size]
        return t_output
```

```
[12]: # A summary function to see the parameters of a module
def summary_nn(mod):
    total_numel = 0
    print(mod)
    for name, param in mod.named_parameters():
        total_numel += param.numel()
        print(f"Param: {name}\tShape: {list(param.shape)}\tNumel: {param.
↪numel()}")
    print(f"Total Numel = {total_numel}")
```

14 Overfitting

- Continuous training will sometimes lead to overfitting
 - the training loss decreases, but the validation loss increases

15 Early stopping

- Training can be stopped when the validation loss is stable for a number of iterations
 - stable means change below a threshold
 - this is to prevent overfitting the training data.
 - we can limit the number of iterations.
- *We are using the loss/accuracy on the (held-out) validation data to estimate the generalization performance of the network*

```
[13]: # Early stopping implementation
class EarlyStopping:
    def __init__(self, patience=5, min_delta=0.0001):
        self.patience = patience
        self.min_delta = min_delta
        self.counter = 0
        self.best_accuracy = 0.0
        self.early_stop = False

    def __call__(self, current_accuracy):
        if current_accuracy - self.best_accuracy > self.min_delta:
            self.best_accuracy = current_accuracy
            self.counter = 0
        else:
            self.counter += 1
            if self.counter >= self.patience:
                self.early_stop = True
        return self.early_stop
```

```
[14]: # define the training loop
def train_model(model, criterion, train_loader, valid_loader, total_epoch: int,
↪patience: int):
```



```

# for early stopping
early_stopping = EarlyStopping(patience=patience)
# record all the losses and accuracies
train_losses = []
train_accs = []
val_losses = []
val_accs = []
# all the epochs
for epoch in range(total_epoch):
    model.train() # set to training mode
    running_loss = 0.0
    correct = 0
    total = 0
    # loop over the training data
    for inputs, labels in train_loader:
        optimizer.zero_grad() # 1. Zero the gradients
        outputs = model(inputs) # 2. Forward pass
        loss = criterion(outputs, labels) # 3. Calculate the loss
        loss.backward() # 4. Backward pass
        optimizer.step() # 5. Optimizer step
        # recordings
        with torch.no_grad():
            running_loss += loss.item()
            predicted = torch.argmax(outputs, -1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
    # results for one epoch
    train_loss = running_loss / len(train_loader)
    train_acc = correct / total
    train_losses.append(train_loss)
    train_accs.append(train_acc)
    # Validation
    model.eval() # set to eval mode
    val_loss = 0.0
    val_correct = 0
    val_total = 0
    with torch.no_grad():
        for inputs, labels in valid_loader:
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            val_loss += loss.item()
            predicted = torch.argmax(outputs, -1)
            val_total += labels.size(0)
            val_correct += (predicted == labels).sum().item()
    val_loss = val_loss / len(valid_loader)
    val_acc = val_correct / val_total
    val_losses.append(val_loss)

```

```

        val_accs.append(val_acc)
        # print results
        print(f'Epoch {epoch+1}: Train Loss: {train_loss:.4f}, Train Acc: {train_acc:.4f}, Val Loss: {val_loss:.4f}, Val Acc: {val_acc:.4f}')
        if early_stopping(val_acc):
            print("Early stopping triggered")
            break
    return {'train_loss': train_losses, 'train_acc': train_accs, 'val_loss': val_losses, 'val_acc': val_accs}

```

```

[15]: # get dataset and dataloader
def get_dataloader(t_input, t_output, batch_size=32, shuffle=False):
    t_input, t_output = torch.as_tensor(t_input), torch.as_tensor(t_output)
    dataset = TensorDataset(t_input, t_output)
    loader = DataLoader(dataset, batch_size=batch_size, shuffle=shuffle)
    return loader

```

- train 1 NN with just one output layer
 - this is the same as logistic regression

```

[16]: model = MyModel(input_size=2, output_size=2, hidden_sizes=[]) # one layer model
criterion = nn.CrossEntropyLoss() # loss function
optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True) # optimizer
train_loader = get_dataloader(trainX, trainY) # data loaders
val_loader = get_dataloader(valX, valY)
# train the model
history = train_model(model, criterion, train_loader, val_loader, total_epoch=100, patience=50)

```

```

Epoch 1: Train Loss: 1.0289, Train Acc: 0.2556, Val Loss: 0.8489, Val Acc: 0.2000
Epoch 2: Train Loss: 0.7875, Train Acc: 0.4000, Val Loss: 0.6500, Val Acc: 0.7000
Epoch 3: Train Loss: 0.5473, Train Acc: 0.7889, Val Loss: 0.5168, Val Acc: 0.8000
Epoch 4: Train Loss: 0.3962, Train Acc: 0.8778, Val Loss: 0.4531, Val Acc: 0.8000
Epoch 5: Train Loss: 0.3205, Train Acc: 0.8889, Val Loss: 0.4296, Val Acc: 0.8000
Epoch 6: Train Loss: 0.2851, Train Acc: 0.8889, Val Loss: 0.4257, Val Acc: 0.8000
Epoch 7: Train Loss: 0.2685, Train Acc: 0.8889, Val Loss: 0.4302, Val Acc: 0.8000
Epoch 8: Train Loss: 0.2607, Train Acc: 0.8889, Val Loss: 0.4375, Val Acc: 0.8000
Epoch 9: Train Loss: 0.2569, Train Acc: 0.8889, Val Loss: 0.4448, Val Acc: 0.8000

```

Epoch 10: Train Loss: 0.2549, Train Acc: 0.8889, Val Loss: 0.4509, Val Acc: 0.8000
Epoch 11: Train Loss: 0.2538, Train Acc: 0.8889, Val Loss: 0.4555, Val Acc: 0.8000
Epoch 12: Train Loss: 0.2530, Train Acc: 0.8889, Val Loss: 0.4586, Val Acc: 0.8000
Epoch 13: Train Loss: 0.2523, Train Acc: 0.8889, Val Loss: 0.4604, Val Acc: 0.8000
Epoch 14: Train Loss: 0.2517, Train Acc: 0.8889, Val Loss: 0.4613, Val Acc: 0.8000
Epoch 15: Train Loss: 0.2510, Train Acc: 0.8889, Val Loss: 0.4616, Val Acc: 0.8000
Epoch 16: Train Loss: 0.2504, Train Acc: 0.8889, Val Loss: 0.4615, Val Acc: 0.8000
Epoch 17: Train Loss: 0.2498, Train Acc: 0.8889, Val Loss: 0.4611, Val Acc: 0.8000
Epoch 18: Train Loss: 0.2493, Train Acc: 0.8889, Val Loss: 0.4608, Val Acc: 0.8000
Epoch 19: Train Loss: 0.2489, Train Acc: 0.9000, Val Loss: 0.4604, Val Acc: 0.8000
Epoch 20: Train Loss: 0.2486, Train Acc: 0.9000, Val Loss: 0.4601, Val Acc: 0.8000
Epoch 21: Train Loss: 0.2483, Train Acc: 0.9000, Val Loss: 0.4598, Val Acc: 0.8000
Epoch 22: Train Loss: 0.2481, Train Acc: 0.8889, Val Loss: 0.4596, Val Acc: 0.8000
Epoch 23: Train Loss: 0.2480, Train Acc: 0.8889, Val Loss: 0.4595, Val Acc: 0.8000
Epoch 24: Train Loss: 0.2479, Train Acc: 0.8889, Val Loss: 0.4594, Val Acc: 0.8000
Epoch 25: Train Loss: 0.2478, Train Acc: 0.8889, Val Loss: 0.4593, Val Acc: 0.8000
Epoch 26: Train Loss: 0.2478, Train Acc: 0.8889, Val Loss: 0.4593, Val Acc: 0.8000
Epoch 27: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4593, Val Acc: 0.8000
Epoch 28: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4593, Val Acc: 0.8000
Epoch 29: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4593, Val Acc: 0.8000
Epoch 30: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4594, Val Acc: 0.8000
Epoch 31: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4594, Val Acc: 0.8000
Epoch 32: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4595, Val Acc: 0.8000
Epoch 33: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4596, Val Acc: 0.8000

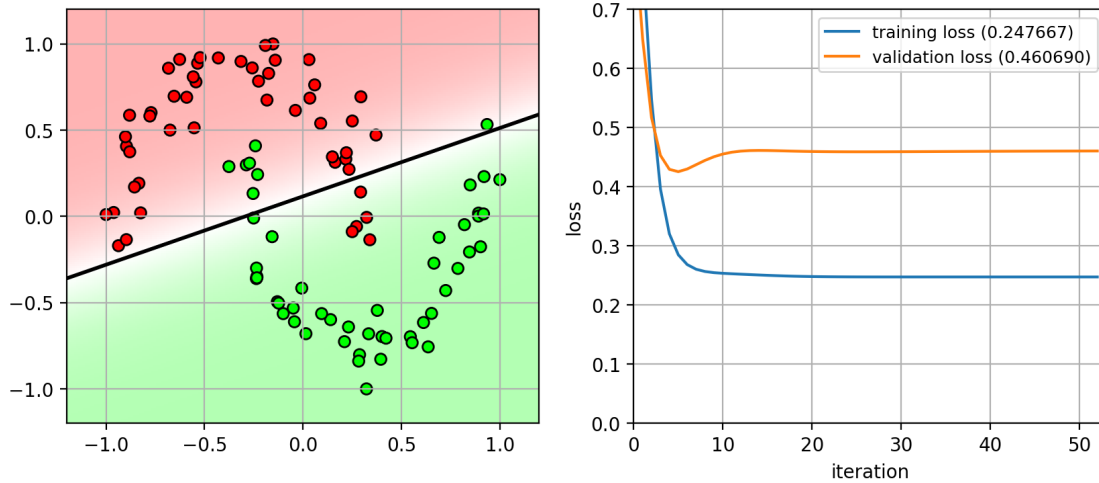
Epoch 34: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4596, Val Acc: 0.8000
Epoch 35: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4597, Val Acc: 0.8000
Epoch 36: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4598, Val Acc: 0.8000
Epoch 37: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4599, Val Acc: 0.8000
Epoch 38: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4599, Val Acc: 0.8000
Epoch 39: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4600, Val Acc: 0.8000
Epoch 40: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4601, Val Acc: 0.8000
Epoch 41: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4601, Val Acc: 0.8000
Epoch 42: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4602, Val Acc: 0.8000
Epoch 43: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4602, Val Acc: 0.8000
Epoch 44: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4603, Val Acc: 0.8000
Epoch 45: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4604, Val Acc: 0.8000
Epoch 46: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4604, Val Acc: 0.8000
Epoch 47: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4605, Val Acc: 0.8000
Epoch 48: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4605, Val Acc: 0.8000
Epoch 49: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4605, Val Acc: 0.8000
Epoch 50: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4606, Val Acc: 0.8000
Epoch 51: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4606, Val Acc: 0.8000
Epoch 52: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4607, Val Acc: 0.8000
Epoch 53: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4607, Val Acc: 0.8000
Early stopping triggered

```
[17]: nnfig = plt.figure(figsize=(10,4))  
      plt.subplot(1,2,1)  
      plot_nn(model, axbox, X, Y)  
      plt.subplot(1,2,2)  
      plot_history(history)
```

```
plt.close()
```

```
nnfig
```

[17]:



- Add one 1 hidden layer with 4 ReLU nodes

```
[18]: model = MyModel(input_size=2, output_size=2, hidden_sizes=[4]) # MLP model
       criterion = nn.CrossEntropyLoss() # loss function
       optimizer = optim.SGD(model.parameters(), lr=0.3, momentum=0.9, nesterov=True)
       ↪ # optimizer
       train_loader = get_dataloader(trainX, trainY) # data loaders
       val_loader = get_dataloader(valX, valY)
       # train the model
       history = train_model(model, criterion, train_loader, val_loader,
       ↪ total_epoch=100, patience=50)
```

```
Epoch 1: Train Loss: 0.6305, Train Acc: 0.5889, Val Loss: 0.5638, Val Acc:
0.7000
Epoch 2: Train Loss: 0.4554, Train Acc: 0.8444, Val Loss: 0.4498, Val Acc:
0.7000
Epoch 3: Train Loss: 0.3019, Train Acc: 0.8667, Val Loss: 0.4678, Val Acc:
0.8000
Epoch 4: Train Loss: 0.2763, Train Acc: 0.8889, Val Loss: 0.5332, Val Acc:
0.8000
Epoch 5: Train Loss: 0.2843, Train Acc: 0.9000, Val Loss: 0.5601, Val Acc:
0.8000
Epoch 6: Train Loss: 0.2938, Train Acc: 0.8778, Val Loss: 0.5586, Val Acc:
0.8000
Epoch 7: Train Loss: 0.2960, Train Acc: 0.8667, Val Loss: 0.5324, Val Acc:
0.8000
Epoch 8: Train Loss: 0.2872, Train Acc: 0.8667, Val Loss: 0.4952, Val Acc:
```

0.8000
Epoch 9: Train Loss: 0.2764, Train Acc: 0.8667, Val Loss: 0.4652, Val Acc:
0.8000
Epoch 10: Train Loss: 0.2709, Train Acc: 0.9000, Val Loss: 0.4473, Val Acc:
0.8000
Epoch 11: Train Loss: 0.2698, Train Acc: 0.9000, Val Loss: 0.4394, Val Acc:
0.8000
Epoch 12: Train Loss: 0.2691, Train Acc: 0.9000, Val Loss: 0.4378, Val Acc:
0.8000
Epoch 13: Train Loss: 0.2673, Train Acc: 0.9000, Val Loss: 0.4395, Val Acc:
0.8000
Epoch 14: Train Loss: 0.2649, Train Acc: 0.9000, Val Loss: 0.4425, Val Acc:
0.8000
Epoch 15: Train Loss: 0.2627, Train Acc: 0.9000, Val Loss: 0.4449, Val Acc:
0.8000
Epoch 16: Train Loss: 0.2610, Train Acc: 0.9000, Val Loss: 0.4443, Val Acc:
0.8000
Epoch 17: Train Loss: 0.2596, Train Acc: 0.9000, Val Loss: 0.4417, Val Acc:
0.8000
Epoch 18: Train Loss: 0.2582, Train Acc: 0.8889, Val Loss: 0.4372, Val Acc:
0.8000
Epoch 19: Train Loss: 0.2566, Train Acc: 0.8889, Val Loss: 0.4312, Val Acc:
0.8000
Epoch 20: Train Loss: 0.2548, Train Acc: 0.8889, Val Loss: 0.4247, Val Acc:
0.8000
Epoch 21: Train Loss: 0.2530, Train Acc: 0.8889, Val Loss: 0.4186, Val Acc:
0.8000
Epoch 22: Train Loss: 0.2512, Train Acc: 0.8889, Val Loss: 0.4129, Val Acc:
0.8000
Epoch 23: Train Loss: 0.2494, Train Acc: 0.8889, Val Loss: 0.4077, Val Acc:
0.8000
Epoch 24: Train Loss: 0.2477, Train Acc: 0.8889, Val Loss: 0.4029, Val Acc:
0.8000
Epoch 25: Train Loss: 0.2460, Train Acc: 0.8889, Val Loss: 0.3985, Val Acc:
0.8000
Epoch 26: Train Loss: 0.2444, Train Acc: 0.8889, Val Loss: 0.3940, Val Acc:
0.8000
Epoch 27: Train Loss: 0.2429, Train Acc: 0.8889, Val Loss: 0.3891, Val Acc:
0.8000
Epoch 28: Train Loss: 0.2416, Train Acc: 0.8889, Val Loss: 0.3842, Val Acc:
0.8000
Epoch 29: Train Loss: 0.2404, Train Acc: 0.8889, Val Loss: 0.3795, Val Acc:
0.8000
Epoch 30: Train Loss: 0.2394, Train Acc: 0.8889, Val Loss: 0.3754, Val Acc:
0.8000
Epoch 31: Train Loss: 0.2385, Train Acc: 0.8889, Val Loss: 0.3721, Val Acc:
0.8000
Epoch 32: Train Loss: 0.2376, Train Acc: 0.8889, Val Loss: 0.3695, Val Acc:

```
0.8000
Epoch 33: Train Loss: 0.2368, Train Acc: 0.8889, Val Loss: 0.3677, Val Acc:
0.8000
Epoch 34: Train Loss: 0.2359, Train Acc: 0.9000, Val Loss: 0.3668, Val Acc:
0.8000
Epoch 35: Train Loss: 0.2344, Train Acc: 0.9000, Val Loss: 0.3640, Val Acc:
0.8000
Epoch 36: Train Loss: 0.2346, Train Acc: 0.9000, Val Loss: 0.3622, Val Acc:
0.8000
Epoch 37: Train Loss: 0.2341, Train Acc: 0.9000, Val Loss: 0.3610, Val Acc:
0.8000
Epoch 38: Train Loss: 0.2329, Train Acc: 0.9000, Val Loss: 0.3585, Val Acc:
0.8000
Epoch 39: Train Loss: 0.2333, Train Acc: 0.9000, Val Loss: 0.3571, Val Acc:
0.8000
Epoch 40: Train Loss: 0.2327, Train Acc: 0.9000, Val Loss: 0.3568, Val Acc:
0.8000
Epoch 41: Train Loss: 0.2324, Train Acc: 0.9000, Val Loss: 0.3554, Val Acc:
0.8000
Epoch 42: Train Loss: 0.2317, Train Acc: 0.9000, Val Loss: 0.3535, Val Acc:
0.8000
Epoch 43: Train Loss: 0.2322, Train Acc: 0.9000, Val Loss: 0.3524, Val Acc:
0.8000
Epoch 44: Train Loss: 0.2317, Train Acc: 0.9000, Val Loss: 0.3523, Val Acc:
0.8000
Epoch 45: Train Loss: 0.2315, Train Acc: 0.9000, Val Loss: 0.3515, Val Acc:
0.8000
Epoch 46: Train Loss: 0.2310, Train Acc: 0.9000, Val Loss: 0.3502, Val Acc:
0.8000
Epoch 47: Train Loss: 0.2311, Train Acc: 0.9000, Val Loss: 0.3500, Val Acc:
0.8000
Epoch 48: Train Loss: 0.2307, Train Acc: 0.9000, Val Loss: 0.3491, Val Acc:
0.8000
Epoch 49: Train Loss: 0.2303, Train Acc: 0.9000, Val Loss: 0.3479, Val Acc:
0.8000
Epoch 50: Train Loss: 0.2313, Train Acc: 0.9000, Val Loss: 0.3473, Val Acc:
0.8000
Epoch 51: Train Loss: 0.2301, Train Acc: 0.9000, Val Loss: 0.3465, Val Acc:
0.8000
Epoch 52: Train Loss: 0.2304, Train Acc: 0.9000, Val Loss: 0.3467, Val Acc:
0.8000
Epoch 53: Train Loss: 0.2304, Train Acc: 0.9000, Val Loss: 0.3468, Val Acc:
0.8000
Early stopping triggered
```

```
[19]: nnfig = plt.figure(figsize=(10,4))
      plt.subplot(1,2,1)
```

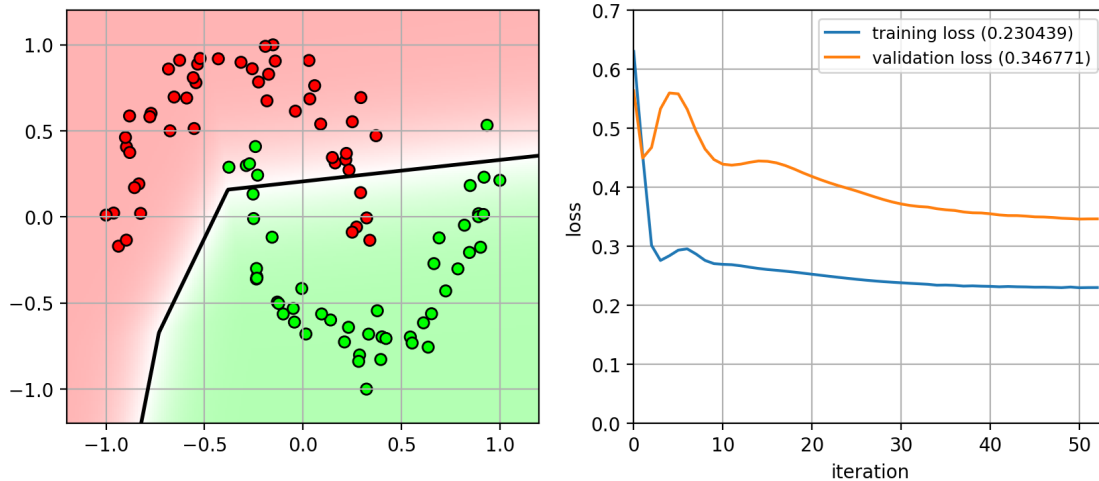
```

plot_nn(model, axbox, X, Y)
plt.subplot(1,2,2)
plot_history(history)
plt.close()

```

nnfig

[19]:



- Let's try more nodes
 - 1 hidden layer with 20 hidden nodes

```

[20]: model = MyModel(input_size=2, output_size=2, hidden_sizes=[20]) # MLP model
criterion = nn.CrossEntropyLoss() # loss function
optimizer = optim.SGD(model.parameters(), lr=0.3, momentum=0.9, nesterov=True)
# optimizer
train_loader = get_dataloader(trainX, trainY) # data loaders
val_loader = get_dataloader(valX, valY)
# train the model
history = train_model(model, criterion, train_loader, val_loader,
    total_epoch=100, patience=50)

```

Epoch 1: Train Loss: 0.6014, Train Acc: 0.6778, Val Loss: 0.4945, Val Acc: 0.7000
 Epoch 2: Train Loss: 0.3375, Train Acc: 0.8889, Val Loss: 0.4680, Val Acc: 0.8000
 Epoch 3: Train Loss: 0.2777, Train Acc: 0.8889, Val Loss: 0.5161, Val Acc: 0.8000
 Epoch 4: Train Loss: 0.2792, Train Acc: 0.9000, Val Loss: 0.5402, Val Acc: 0.8000
 Epoch 5: Train Loss: 0.2866, Train Acc: 0.8778, Val Loss: 0.5418, Val Acc: 0.8000
 Epoch 6: Train Loss: 0.2877, Train Acc: 0.8667, Val Loss: 0.5130, Val Acc:

0.8000
Epoch 7: Train Loss: 0.2744, Train Acc: 0.8778, Val Loss: 0.4772, Val Acc:
0.8000
Epoch 8: Train Loss: 0.2636, Train Acc: 0.8778, Val Loss: 0.4544, Val Acc:
0.8000
Epoch 9: Train Loss: 0.2595, Train Acc: 0.8889, Val Loss: 0.4366, Val Acc:
0.8000
Epoch 10: Train Loss: 0.2569, Train Acc: 0.8778, Val Loss: 0.4291, Val Acc:
0.8000
Epoch 11: Train Loss: 0.2505, Train Acc: 0.8889, Val Loss: 0.4258, Val Acc:
0.8000
Epoch 12: Train Loss: 0.2425, Train Acc: 0.8889, Val Loss: 0.4277, Val Acc:
0.8000
Epoch 13: Train Loss: 0.2352, Train Acc: 0.8778, Val Loss: 0.4295, Val Acc:
0.8000
Epoch 14: Train Loss: 0.2299, Train Acc: 0.8778, Val Loss: 0.4142, Val Acc:
0.8000
Epoch 15: Train Loss: 0.2241, Train Acc: 0.8778, Val Loss: 0.3980, Val Acc:
0.8000
Epoch 16: Train Loss: 0.2162, Train Acc: 0.8889, Val Loss: 0.3850, Val Acc:
0.8000
Epoch 17: Train Loss: 0.2090, Train Acc: 0.9000, Val Loss: 0.3621, Val Acc:
0.8000
Epoch 18: Train Loss: 0.1997, Train Acc: 0.9000, Val Loss: 0.3441, Val Acc:
0.8000
Epoch 19: Train Loss: 0.1893, Train Acc: 0.9000, Val Loss: 0.3268, Val Acc:
0.8000
Epoch 20: Train Loss: 0.1789, Train Acc: 0.9111, Val Loss: 0.3062, Val Acc:
0.8000
Epoch 21: Train Loss: 0.1670, Train Acc: 0.9222, Val Loss: 0.2840, Val Acc:
0.8000
Epoch 22: Train Loss: 0.1548, Train Acc: 0.9222, Val Loss: 0.2639, Val Acc:
0.8000
Epoch 23: Train Loss: 0.1425, Train Acc: 0.9222, Val Loss: 0.2403, Val Acc:
0.9000
Epoch 24: Train Loss: 0.1294, Train Acc: 0.9222, Val Loss: 0.2190, Val Acc:
0.9000
Epoch 25: Train Loss: 0.1167, Train Acc: 0.9333, Val Loss: 0.1999, Val Acc:
0.9000
Epoch 26: Train Loss: 0.1048, Train Acc: 0.9444, Val Loss: 0.1847, Val Acc:
0.9000
Epoch 27: Train Loss: 0.0937, Train Acc: 0.9444, Val Loss: 0.1695, Val Acc:
0.9000
Epoch 28: Train Loss: 0.0833, Train Acc: 0.9889, Val Loss: 0.1537, Val Acc:
0.9000
Epoch 29: Train Loss: 0.0739, Train Acc: 0.9889, Val Loss: 0.1400, Val Acc:
0.9000
Epoch 30: Train Loss: 0.0655, Train Acc: 0.9889, Val Loss: 0.1291, Val Acc:

1.0000
Epoch 31: Train Loss: 0.0582, Train Acc: 1.0000, Val Loss: 0.1214, Val Acc: 1.0000
Epoch 32: Train Loss: 0.0522, Train Acc: 1.0000, Val Loss: 0.1143, Val Acc: 1.0000
Epoch 33: Train Loss: 0.0465, Train Acc: 1.0000, Val Loss: 0.1048, Val Acc: 1.0000
Epoch 34: Train Loss: 0.0420, Train Acc: 1.0000, Val Loss: 0.0985, Val Acc: 1.0000
Epoch 35: Train Loss: 0.0379, Train Acc: 1.0000, Val Loss: 0.0928, Val Acc: 1.0000
Epoch 36: Train Loss: 0.0344, Train Acc: 1.0000, Val Loss: 0.0880, Val Acc: 1.0000
Epoch 37: Train Loss: 0.0314, Train Acc: 1.0000, Val Loss: 0.0837, Val Acc: 1.0000
Epoch 38: Train Loss: 0.0289, Train Acc: 1.0000, Val Loss: 0.0780, Val Acc: 1.0000
Epoch 39: Train Loss: 0.0268, Train Acc: 1.0000, Val Loss: 0.0756, Val Acc: 1.0000
Epoch 40: Train Loss: 0.0247, Train Acc: 1.0000, Val Loss: 0.0720, Val Acc: 1.0000
Epoch 41: Train Loss: 0.0230, Train Acc: 1.0000, Val Loss: 0.0696, Val Acc: 1.0000
Epoch 42: Train Loss: 0.0214, Train Acc: 1.0000, Val Loss: 0.0660, Val Acc: 1.0000
Epoch 43: Train Loss: 0.0201, Train Acc: 1.0000, Val Loss: 0.0647, Val Acc: 1.0000
Epoch 44: Train Loss: 0.0189, Train Acc: 1.0000, Val Loss: 0.0622, Val Acc: 1.0000
Epoch 45: Train Loss: 0.0178, Train Acc: 1.0000, Val Loss: 0.0594, Val Acc: 1.0000
Epoch 46: Train Loss: 0.0168, Train Acc: 1.0000, Val Loss: 0.0577, Val Acc: 1.0000
Epoch 47: Train Loss: 0.0160, Train Acc: 1.0000, Val Loss: 0.0563, Val Acc: 1.0000
Epoch 48: Train Loss: 0.0151, Train Acc: 1.0000, Val Loss: 0.0556, Val Acc: 1.0000
Epoch 49: Train Loss: 0.0144, Train Acc: 1.0000, Val Loss: 0.0533, Val Acc: 1.0000
Epoch 50: Train Loss: 0.0137, Train Acc: 1.0000, Val Loss: 0.0521, Val Acc: 1.0000
Epoch 51: Train Loss: 0.0132, Train Acc: 1.0000, Val Loss: 0.0513, Val Acc: 1.0000
Epoch 52: Train Loss: 0.0126, Train Acc: 1.0000, Val Loss: 0.0502, Val Acc: 1.0000
Epoch 53: Train Loss: 0.0120, Train Acc: 1.0000, Val Loss: 0.0482, Val Acc: 1.0000
Epoch 54: Train Loss: 0.0116, Train Acc: 1.0000, Val Loss: 0.0484, Val Acc:

1.0000
Epoch 55: Train Loss: 0.0111, Train Acc: 1.0000, Val Loss: 0.0468, Val Acc: 1.0000
Epoch 56: Train Loss: 0.0107, Train Acc: 1.0000, Val Loss: 0.0468, Val Acc: 1.0000
Epoch 57: Train Loss: 0.0103, Train Acc: 1.0000, Val Loss: 0.0453, Val Acc: 1.0000
Epoch 58: Train Loss: 0.0099, Train Acc: 1.0000, Val Loss: 0.0445, Val Acc: 1.0000
Epoch 59: Train Loss: 0.0096, Train Acc: 1.0000, Val Loss: 0.0438, Val Acc: 1.0000
Epoch 60: Train Loss: 0.0093, Train Acc: 1.0000, Val Loss: 0.0431, Val Acc: 1.0000
Epoch 61: Train Loss: 0.0090, Train Acc: 1.0000, Val Loss: 0.0421, Val Acc: 1.0000
Epoch 62: Train Loss: 0.0087, Train Acc: 1.0000, Val Loss: 0.0414, Val Acc: 1.0000
Epoch 63: Train Loss: 0.0084, Train Acc: 1.0000, Val Loss: 0.0402, Val Acc: 1.0000
Epoch 64: Train Loss: 0.0082, Train Acc: 1.0000, Val Loss: 0.0397, Val Acc: 1.0000
Epoch 65: Train Loss: 0.0080, Train Acc: 1.0000, Val Loss: 0.0391, Val Acc: 1.0000
Epoch 66: Train Loss: 0.0077, Train Acc: 1.0000, Val Loss: 0.0384, Val Acc: 1.0000
Epoch 67: Train Loss: 0.0075, Train Acc: 1.0000, Val Loss: 0.0377, Val Acc: 1.0000
Epoch 68: Train Loss: 0.0073, Train Acc: 1.0000, Val Loss: 0.0374, Val Acc: 1.0000
Epoch 69: Train Loss: 0.0071, Train Acc: 1.0000, Val Loss: 0.0370, Val Acc: 1.0000
Epoch 70: Train Loss: 0.0069, Train Acc: 1.0000, Val Loss: 0.0362, Val Acc: 1.0000
Epoch 71: Train Loss: 0.0068, Train Acc: 1.0000, Val Loss: 0.0359, Val Acc: 1.0000
Epoch 72: Train Loss: 0.0066, Train Acc: 1.0000, Val Loss: 0.0352, Val Acc: 1.0000
Epoch 73: Train Loss: 0.0064, Train Acc: 1.0000, Val Loss: 0.0350, Val Acc: 1.0000
Epoch 74: Train Loss: 0.0063, Train Acc: 1.0000, Val Loss: 0.0342, Val Acc: 1.0000
Epoch 75: Train Loss: 0.0061, Train Acc: 1.0000, Val Loss: 0.0338, Val Acc: 1.0000
Epoch 76: Train Loss: 0.0060, Train Acc: 1.0000, Val Loss: 0.0336, Val Acc: 1.0000
Epoch 77: Train Loss: 0.0059, Train Acc: 1.0000, Val Loss: 0.0331, Val Acc: 1.0000
Epoch 78: Train Loss: 0.0057, Train Acc: 1.0000, Val Loss: 0.0325, Val Acc:

```

1.0000
Epoch 79: Train Loss: 0.0056, Train Acc: 1.0000, Val Loss: 0.0322, Val Acc:
1.0000
Epoch 80: Train Loss: 0.0055, Train Acc: 1.0000, Val Loss: 0.0320, Val Acc:
1.0000
Early stopping triggered

```

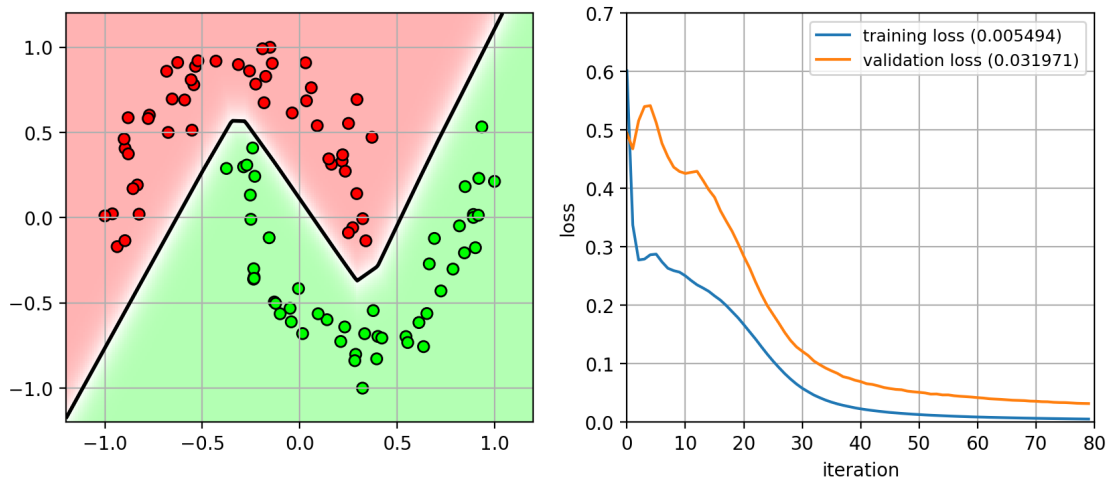
```

[21]: nnfig = plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plot_nn(model, axbox, X, Y)
plt.subplot(1,2,2)
plot_history(history)
plt.close()

nnfig

```

[21]:



16 Universal Approximation Theorem

- Cybenko (1989), Hornik (1991)
 - A multi-layer perceptron with a single hidden layer and a finite number of nodes can approximate any continuous function up to a desired error.
 - * The number of nodes needed might be very large.
 - * Doesn't say anything about how difficult it is to train it.
- How many hidden nodes are needed?
 - In the worst case, consider binary functions mapping $\{0,1\}^n \rightarrow \{0,1\}$
 - * there are 2^n possible inputs.
 - * there are 2^{2^n} possible functions (each input has 2 possible outputs)
 - * thus, we need 2^n bits in the hidden layer to select one of these functions.
 - need $O(2^n)$ nodes in the hidden layer, exponential in the size of the input!

- *How to train this model?*
 - According to the “no free lunch theorem”, there is no universally best learning algorithm!
- Deep learning corollary
 - *The number of functions representable by a deep network requires an exponential number of nodes for a shallow network with 1 hidden-layer.*
 - A deep network can learn the same function using less nodes.
 - Given the same number of nodes, a deep network can learn more complex functions.
- Doesn't say anything about how difficult it is to train it.

17 Example

- Network with 1 hidden layer
 - input (2D) -> 40 hidden nodes -> output (2D)

```
[22]: trainX2 = vstack((trainX, trainX+array([1.2,0], dtype=np.float32)))
      valX2 = vstack((valX, valX+array([1.2,0], dtype=np.float32)))
      trainY2 = r_[trainY, trainY]
      valY2 = r_[valY, valY]
      axbox2 = [-1.2, 2.5, -1.2, 1.2]
```

```
[23]: model = MyModel(input_size=2, output_size=2, hidden_sizes=[40]) # MLP model
      criterion = nn.CrossEntropyLoss() # loss function
      optimizer = optim.SGD(model.parameters(), lr=0.2, momentum=0.9, nesterov=True) ↵
      ↪# optimizer
      train_loader = get_dataloader(trainX2, trainY2) # data loaders
      val_loader = get_dataloader(valX2, valY2)
      # train the model
      history = train_model(model, criterion, train_loader, val_loader, ↵
      ↪total_epoch=100, patience=50)
```

```
Epoch 1: Train Loss: 0.5441, Train Acc: 0.6833, Val Loss: 0.4883, Val Acc:
0.8000
Epoch 2: Train Loss: 0.3665, Train Acc: 0.8500, Val Loss: 0.5378, Val Acc:
0.8000
Epoch 3: Train Loss: 0.3475, Train Acc: 0.8500, Val Loss: 0.5654, Val Acc:
0.7000
Epoch 4: Train Loss: 0.3257, Train Acc: 0.8444, Val Loss: 0.4911, Val Acc:
0.8500
Epoch 5: Train Loss: 0.3034, Train Acc: 0.8556, Val Loss: 0.4588, Val Acc:
0.8000
Epoch 6: Train Loss: 0.3019, Train Acc: 0.8611, Val Loss: 0.4507, Val Acc:
0.8000
Epoch 7: Train Loss: 0.2926, Train Acc: 0.8556, Val Loss: 0.4476, Val Acc:
0.8000
Epoch 8: Train Loss: 0.2890, Train Acc: 0.8500, Val Loss: 0.4448, Val Acc:
0.8500
Epoch 9: Train Loss: 0.2840, Train Acc: 0.8500, Val Loss: 0.4432, Val Acc:
```

0.8500
Epoch 10: Train Loss: 0.2790, Train Acc: 0.8556, Val Loss: 0.4415, Val Acc:
0.8500
Epoch 11: Train Loss: 0.2750, Train Acc: 0.8667, Val Loss: 0.4326, Val Acc:
0.8000
Epoch 12: Train Loss: 0.2723, Train Acc: 0.8667, Val Loss: 0.4278, Val Acc:
0.8500
Epoch 13: Train Loss: 0.2678, Train Acc: 0.8778, Val Loss: 0.4207, Val Acc:
0.8500
Epoch 14: Train Loss: 0.2659, Train Acc: 0.8722, Val Loss: 0.4108, Val Acc:
0.8500
Epoch 15: Train Loss: 0.2627, Train Acc: 0.8778, Val Loss: 0.4045, Val Acc:
0.8500
Epoch 16: Train Loss: 0.2542, Train Acc: 0.8833, Val Loss: 0.3963, Val Acc:
0.8500
Epoch 17: Train Loss: 0.2510, Train Acc: 0.8889, Val Loss: 0.3880, Val Acc:
0.8500
Epoch 18: Train Loss: 0.2481, Train Acc: 0.8833, Val Loss: 0.3753, Val Acc:
0.8500
Epoch 19: Train Loss: 0.2420, Train Acc: 0.8944, Val Loss: 0.3640, Val Acc:
0.8500
Epoch 20: Train Loss: 0.2331, Train Acc: 0.8944, Val Loss: 0.3560, Val Acc:
0.8500
Epoch 21: Train Loss: 0.2285, Train Acc: 0.8944, Val Loss: 0.3449, Val Acc:
0.8500
Epoch 22: Train Loss: 0.2246, Train Acc: 0.8944, Val Loss: 0.3369, Val Acc:
0.8500
Epoch 23: Train Loss: 0.2188, Train Acc: 0.8944, Val Loss: 0.3271, Val Acc:
0.8500
Epoch 24: Train Loss: 0.2116, Train Acc: 0.9000, Val Loss: 0.3139, Val Acc:
0.8500
Epoch 25: Train Loss: 0.2049, Train Acc: 0.9111, Val Loss: 0.3062, Val Acc:
0.8500
Epoch 26: Train Loss: 0.1993, Train Acc: 0.9222, Val Loss: 0.3013, Val Acc:
0.8500
Epoch 27: Train Loss: 0.1986, Train Acc: 0.9167, Val Loss: 0.2841, Val Acc:
0.8500
Epoch 28: Train Loss: 0.1859, Train Acc: 0.9278, Val Loss: 0.2878, Val Acc:
0.8500
Epoch 29: Train Loss: 0.1836, Train Acc: 0.9333, Val Loss: 0.2623, Val Acc:
0.8500
Epoch 30: Train Loss: 0.1763, Train Acc: 0.9444, Val Loss: 0.2615, Val Acc:
0.8500
Epoch 31: Train Loss: 0.1794, Train Acc: 0.9333, Val Loss: 0.2476, Val Acc:
0.8500
Epoch 32: Train Loss: 0.1616, Train Acc: 0.9444, Val Loss: 0.2437, Val Acc:
0.8500
Epoch 33: Train Loss: 0.1550, Train Acc: 0.9389, Val Loss: 0.2418, Val Acc:

0.8500
Epoch 34: Train Loss: 0.1527, Train Acc: 0.9444, Val Loss: 0.2272, Val Acc: 0.8500
Epoch 35: Train Loss: 0.1490, Train Acc: 0.9444, Val Loss: 0.2275, Val Acc: 0.8500
Epoch 36: Train Loss: 0.1496, Train Acc: 0.9556, Val Loss: 0.1927, Val Acc: 0.8500
Epoch 37: Train Loss: 0.1474, Train Acc: 0.9500, Val Loss: 0.2211, Val Acc: 0.8500
Epoch 38: Train Loss: 0.1420, Train Acc: 0.9500, Val Loss: 0.1940, Val Acc: 0.8500
Epoch 39: Train Loss: 0.1324, Train Acc: 0.9500, Val Loss: 0.2188, Val Acc: 0.8500
Epoch 40: Train Loss: 0.1281, Train Acc: 0.9611, Val Loss: 0.1913, Val Acc: 0.8500
Epoch 41: Train Loss: 0.1177, Train Acc: 0.9500, Val Loss: 0.1919, Val Acc: 0.8500
Epoch 42: Train Loss: 0.1170, Train Acc: 0.9611, Val Loss: 0.1786, Val Acc: 0.8500
Epoch 43: Train Loss: 0.1148, Train Acc: 0.9667, Val Loss: 0.1924, Val Acc: 0.8500
Epoch 44: Train Loss: 0.1164, Train Acc: 0.9500, Val Loss: 0.1698, Val Acc: 0.9000
Epoch 45: Train Loss: 0.1132, Train Acc: 0.9667, Val Loss: 0.1823, Val Acc: 0.8500
Epoch 46: Train Loss: 0.1082, Train Acc: 0.9611, Val Loss: 0.1645, Val Acc: 0.9000
Epoch 47: Train Loss: 0.1027, Train Acc: 0.9667, Val Loss: 0.1727, Val Acc: 0.9000
Epoch 48: Train Loss: 0.0980, Train Acc: 0.9611, Val Loss: 0.1554, Val Acc: 0.9000
Epoch 49: Train Loss: 0.0921, Train Acc: 0.9667, Val Loss: 0.1589, Val Acc: 0.9000
Epoch 50: Train Loss: 0.0866, Train Acc: 0.9611, Val Loss: 0.1451, Val Acc: 0.9000
Epoch 51: Train Loss: 0.0814, Train Acc: 0.9667, Val Loss: 0.1501, Val Acc: 0.9500
Epoch 52: Train Loss: 0.0811, Train Acc: 0.9667, Val Loss: 0.1364, Val Acc: 0.9000
Epoch 53: Train Loss: 0.0782, Train Acc: 0.9667, Val Loss: 0.1457, Val Acc: 0.9500
Epoch 54: Train Loss: 0.0778, Train Acc: 0.9667, Val Loss: 0.1277, Val Acc: 0.9500
Epoch 55: Train Loss: 0.0746, Train Acc: 0.9667, Val Loss: 0.1297, Val Acc: 0.9500
Epoch 56: Train Loss: 0.0708, Train Acc: 0.9667, Val Loss: 0.1173, Val Acc: 1.0000
Epoch 57: Train Loss: 0.0657, Train Acc: 0.9833, Val Loss: 0.1244, Val Acc:

0.9500

Epoch 58: Train Loss: 0.0636, Train Acc: 0.9833, Val Loss: 0.1091, Val Acc: 1.0000

Epoch 59: Train Loss: 0.0621, Train Acc: 0.9833, Val Loss: 0.1256, Val Acc: 0.9500

Epoch 60: Train Loss: 0.0604, Train Acc: 0.9833, Val Loss: 0.1043, Val Acc: 1.0000

Epoch 61: Train Loss: 0.0551, Train Acc: 0.9833, Val Loss: 0.1151, Val Acc: 0.9500

Epoch 62: Train Loss: 0.0557, Train Acc: 0.9889, Val Loss: 0.1010, Val Acc: 1.0000

Epoch 63: Train Loss: 0.0528, Train Acc: 0.9889, Val Loss: 0.1089, Val Acc: 0.9500

Epoch 64: Train Loss: 0.0501, Train Acc: 0.9889, Val Loss: 0.0907, Val Acc: 1.0000

Epoch 65: Train Loss: 0.0499, Train Acc: 1.0000, Val Loss: 0.1084, Val Acc: 0.9500

Epoch 66: Train Loss: 0.0511, Train Acc: 0.9889, Val Loss: 0.0923, Val Acc: 1.0000

Epoch 67: Train Loss: 0.0465, Train Acc: 0.9944, Val Loss: 0.1013, Val Acc: 0.9500

Epoch 68: Train Loss: 0.0442, Train Acc: 0.9889, Val Loss: 0.0850, Val Acc: 1.0000

Epoch 69: Train Loss: 0.0414, Train Acc: 0.9889, Val Loss: 0.0860, Val Acc: 1.0000

Epoch 70: Train Loss: 0.0409, Train Acc: 1.0000, Val Loss: 0.0963, Val Acc: 0.9500

Epoch 71: Train Loss: 0.0389, Train Acc: 0.9944, Val Loss: 0.0756, Val Acc: 1.0000

Epoch 72: Train Loss: 0.0365, Train Acc: 1.0000, Val Loss: 0.0870, Val Acc: 0.9500

Epoch 73: Train Loss: 0.0363, Train Acc: 1.0000, Val Loss: 0.0738, Val Acc: 1.0000

Epoch 74: Train Loss: 0.0337, Train Acc: 1.0000, Val Loss: 0.0815, Val Acc: 1.0000

Epoch 75: Train Loss: 0.0322, Train Acc: 1.0000, Val Loss: 0.0675, Val Acc: 1.0000

Epoch 76: Train Loss: 0.0313, Train Acc: 1.0000, Val Loss: 0.0747, Val Acc: 1.0000

Epoch 77: Train Loss: 0.0317, Train Acc: 1.0000, Val Loss: 0.0657, Val Acc: 1.0000

Epoch 78: Train Loss: 0.0298, Train Acc: 1.0000, Val Loss: 0.0775, Val Acc: 1.0000

Epoch 79: Train Loss: 0.0287, Train Acc: 1.0000, Val Loss: 0.0607, Val Acc: 1.0000

Epoch 80: Train Loss: 0.0276, Train Acc: 1.0000, Val Loss: 0.0711, Val Acc: 1.0000

Epoch 81: Train Loss: 0.0276, Train Acc: 1.0000, Val Loss: 0.0619, Val Acc:


```

1.0000
Epoch 82: Train Loss: 0.0258, Train Acc: 1.0000, Val Loss: 0.0654, Val Acc:
1.0000
Epoch 83: Train Loss: 0.0248, Train Acc: 1.0000, Val Loss: 0.0615, Val Acc:
1.0000
Epoch 84: Train Loss: 0.0235, Train Acc: 1.0000, Val Loss: 0.0553, Val Acc:
1.0000
Epoch 85: Train Loss: 0.0243, Train Acc: 1.0000, Val Loss: 0.0607, Val Acc:
1.0000
Epoch 86: Train Loss: 0.0225, Train Acc: 1.0000, Val Loss: 0.0557, Val Acc:
1.0000
Epoch 87: Train Loss: 0.0217, Train Acc: 1.0000, Val Loss: 0.0572, Val Acc:
1.0000
Epoch 88: Train Loss: 0.0209, Train Acc: 1.0000, Val Loss: 0.0522, Val Acc:
1.0000
Epoch 89: Train Loss: 0.0214, Train Acc: 1.0000, Val Loss: 0.0527, Val Acc:
1.0000
Epoch 90: Train Loss: 0.0199, Train Acc: 1.0000, Val Loss: 0.0553, Val Acc:
1.0000
Epoch 91: Train Loss: 0.0193, Train Acc: 1.0000, Val Loss: 0.0505, Val Acc:
1.0000
Epoch 92: Train Loss: 0.0187, Train Acc: 1.0000, Val Loss: 0.0506, Val Acc:
1.0000
Epoch 93: Train Loss: 0.0191, Train Acc: 1.0000, Val Loss: 0.0491, Val Acc:
1.0000
Epoch 94: Train Loss: 0.0179, Train Acc: 1.0000, Val Loss: 0.0501, Val Acc:
1.0000
Epoch 95: Train Loss: 0.0171, Train Acc: 1.0000, Val Loss: 0.0483, Val Acc:
1.0000
Epoch 96: Train Loss: 0.0166, Train Acc: 1.0000, Val Loss: 0.0457, Val Acc:
1.0000
Epoch 97: Train Loss: 0.0171, Train Acc: 1.0000, Val Loss: 0.0461, Val Acc:
1.0000
Epoch 98: Train Loss: 0.0163, Train Acc: 1.0000, Val Loss: 0.0472, Val Acc:
1.0000
Epoch 99: Train Loss: 0.0155, Train Acc: 1.0000, Val Loss: 0.0448, Val Acc:
1.0000
Epoch 100: Train Loss: 0.0153, Train Acc: 1.0000, Val Loss: 0.0431, Val Acc:
1.0000

```

```
[24]: summary_nn(model)
```

```

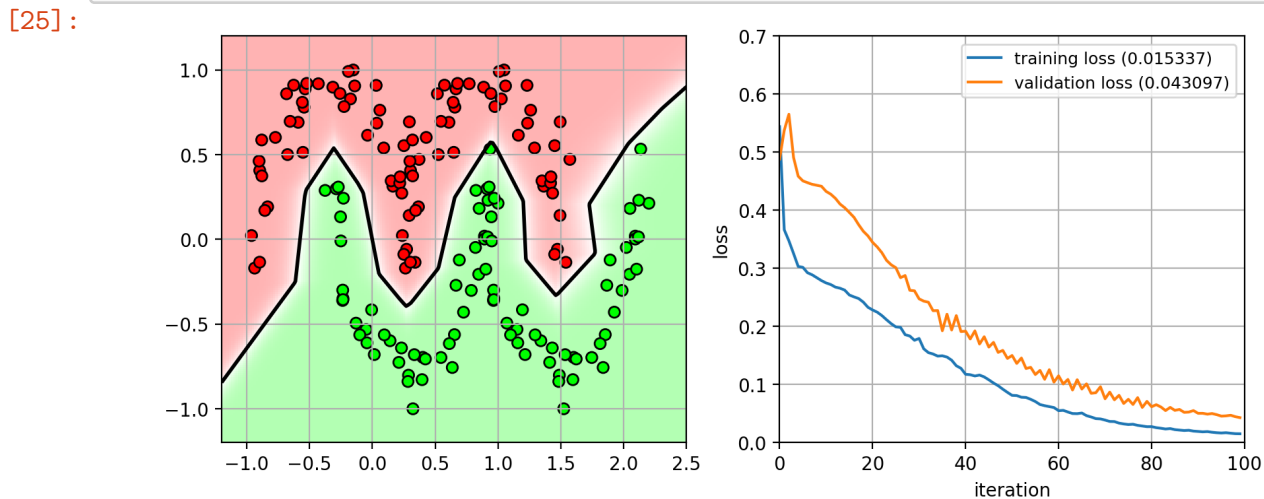
MyModel(
  (network): Sequential(
    (0): Linear(in_features=2, out_features=40, bias=True)
    (1): ReLU()
    (2): Linear(in_features=40, out_features=2, bias=True)
  )

```

```
)
Param: network.0.weight Shape: [40, 2] Numel: 80
Param: network.0.bias Shape: [40] Numel: 40
Param: network.2.weight Shape: [2, 40] Numel: 80
Param: network.2.bias Shape: [2] Numel: 2
Total Numel = 202
```

```
[25]: nnfig = plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plot_nn(model, axbox2, trainX2, trainY2)
plt.subplot(1,2,2)
plot_history(history)
plt.close()

nnfig
```



- More hidden layers:
 - input (2D) -> 16 nodes -> 16 nodes -> output (2D)

```
[26]: model = MyModel(input_size=2, output_size=2, hidden_sizes=[16, 16]) # MLP model
criterion = nn.CrossEntropyLoss() # loss function
optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True) ↵
↵# optimizer
train_loader = get_dataloader(trainX2, trainY2) # data loaders
val_loader = get_dataloader(valX2, valY2)
# train the model
history = train_model(model, criterion, train_loader, val_loader, ↵
↵total_epoch=100, patience=50)
```

Epoch 1: Train Loss: 0.6153, Train Acc: 0.7389, Val Loss: 0.5550, Val Acc: 0.7000

Epoch 2: Train Loss: 0.3982, Train Acc: 0.8667, Val Loss: 0.4936, Val Acc:

0.8000
Epoch 3: Train Loss: 0.3542, Train Acc: 0.8500, Val Loss: 0.5445, Val Acc:
0.8000
Epoch 4: Train Loss: 0.3557, Train Acc: 0.8611, Val Loss: 0.5140, Val Acc:
0.8500
Epoch 5: Train Loss: 0.3206, Train Acc: 0.8333, Val Loss: 0.4709, Val Acc:
0.8000
Epoch 6: Train Loss: 0.3078, Train Acc: 0.8444, Val Loss: 0.4589, Val Acc:
0.8000
Epoch 7: Train Loss: 0.3065, Train Acc: 0.8611, Val Loss: 0.4617, Val Acc:
0.8000
Epoch 8: Train Loss: 0.3043, Train Acc: 0.8500, Val Loss: 0.4640, Val Acc:
0.8000
Epoch 9: Train Loss: 0.3011, Train Acc: 0.8444, Val Loss: 0.4645, Val Acc:
0.8000
Epoch 10: Train Loss: 0.2975, Train Acc: 0.8389, Val Loss: 0.4613, Val Acc:
0.8500
Epoch 11: Train Loss: 0.2940, Train Acc: 0.8500, Val Loss: 0.4555, Val Acc:
0.8500
Epoch 12: Train Loss: 0.2913, Train Acc: 0.8500, Val Loss: 0.4516, Val Acc:
0.8500
Epoch 13: Train Loss: 0.2892, Train Acc: 0.8500, Val Loss: 0.4490, Val Acc:
0.8500
Epoch 14: Train Loss: 0.2861, Train Acc: 0.8500, Val Loss: 0.4480, Val Acc:
0.8500
Epoch 15: Train Loss: 0.2833, Train Acc: 0.8667, Val Loss: 0.4436, Val Acc:
0.8500
Epoch 16: Train Loss: 0.2803, Train Acc: 0.8722, Val Loss: 0.4382, Val Acc:
0.8500
Epoch 17: Train Loss: 0.2780, Train Acc: 0.8667, Val Loss: 0.4303, Val Acc:
0.8500
Epoch 18: Train Loss: 0.2740, Train Acc: 0.8667, Val Loss: 0.4248, Val Acc:
0.8500
Epoch 19: Train Loss: 0.2703, Train Acc: 0.8667, Val Loss: 0.4177, Val Acc:
0.8500
Epoch 20: Train Loss: 0.2658, Train Acc: 0.8722, Val Loss: 0.4114, Val Acc:
0.8500
Epoch 21: Train Loss: 0.2616, Train Acc: 0.8778, Val Loss: 0.4027, Val Acc:
0.8500
Epoch 22: Train Loss: 0.2565, Train Acc: 0.8778, Val Loss: 0.3955, Val Acc:
0.8500
Epoch 23: Train Loss: 0.2524, Train Acc: 0.8722, Val Loss: 0.3828, Val Acc:
0.8500
Epoch 24: Train Loss: 0.2458, Train Acc: 0.8778, Val Loss: 0.3687, Val Acc:
0.8500
Epoch 25: Train Loss: 0.2397, Train Acc: 0.8778, Val Loss: 0.3604, Val Acc:
0.8500
Epoch 26: Train Loss: 0.2357, Train Acc: 0.8833, Val Loss: 0.3474, Val Acc:

0.8500
Epoch 27: Train Loss: 0.2303, Train Acc: 0.8778, Val Loss: 0.3238, Val Acc: 0.8500
Epoch 28: Train Loss: 0.2191, Train Acc: 0.8833, Val Loss: 0.3161, Val Acc: 0.8500
Epoch 29: Train Loss: 0.2176, Train Acc: 0.8944, Val Loss: 0.3041, Val Acc: 0.8500
Epoch 30: Train Loss: 0.2080, Train Acc: 0.9056, Val Loss: 0.3067, Val Acc: 0.8500
Epoch 31: Train Loss: 0.1975, Train Acc: 0.9056, Val Loss: 0.2725, Val Acc: 0.8500
Epoch 32: Train Loss: 0.1815, Train Acc: 0.9278, Val Loss: 0.2712, Val Acc: 0.8500
Epoch 33: Train Loss: 0.1762, Train Acc: 0.9278, Val Loss: 0.2531, Val Acc: 0.8500
Epoch 34: Train Loss: 0.1561, Train Acc: 0.9389, Val Loss: 0.2205, Val Acc: 0.8500
Epoch 35: Train Loss: 0.1444, Train Acc: 0.9444, Val Loss: 0.2262, Val Acc: 0.9000
Epoch 36: Train Loss: 0.1499, Train Acc: 0.9500, Val Loss: 0.2010, Val Acc: 0.9000
Epoch 37: Train Loss: 0.1176, Train Acc: 0.9556, Val Loss: 0.2529, Val Acc: 0.8500
Epoch 38: Train Loss: 0.1538, Train Acc: 0.9389, Val Loss: 0.1754, Val Acc: 0.9000
Epoch 39: Train Loss: 0.1062, Train Acc: 0.9611, Val Loss: 0.1505, Val Acc: 0.9500
Epoch 40: Train Loss: 0.0771, Train Acc: 0.9889, Val Loss: 0.1770, Val Acc: 0.9500
Epoch 41: Train Loss: 0.0733, Train Acc: 0.9889, Val Loss: 0.0944, Val Acc: 1.0000
Epoch 42: Train Loss: 0.0519, Train Acc: 0.9889, Val Loss: 0.1233, Val Acc: 0.9500
Epoch 43: Train Loss: 0.0492, Train Acc: 0.9889, Val Loss: 0.0872, Val Acc: 1.0000
Epoch 44: Train Loss: 0.0387, Train Acc: 0.9944, Val Loss: 0.1281, Val Acc: 0.9500
Epoch 45: Train Loss: 0.0504, Train Acc: 0.9889, Val Loss: 0.0637, Val Acc: 1.0000
Epoch 46: Train Loss: 0.0286, Train Acc: 1.0000, Val Loss: 0.1044, Val Acc: 0.9500
Epoch 47: Train Loss: 0.0341, Train Acc: 0.9889, Val Loss: 0.0607, Val Acc: 1.0000
Epoch 48: Train Loss: 0.0287, Train Acc: 1.0000, Val Loss: 0.1044, Val Acc: 0.9500
Epoch 49: Train Loss: 0.0326, Train Acc: 0.9889, Val Loss: 0.0582, Val Acc: 1.0000
Epoch 50: Train Loss: 0.0333, Train Acc: 0.9944, Val Loss: 0.0962, Val Acc:

0.9500

Epoch 51: Train Loss: 0.0288, Train Acc: 0.9889, Val Loss: 0.0543, Val Acc: 1.0000

Epoch 52: Train Loss: 0.0170, Train Acc: 1.0000, Val Loss: 0.0560, Val Acc: 1.0000

Epoch 53: Train Loss: 0.0169, Train Acc: 0.9889, Val Loss: 0.0407, Val Acc: 1.0000

Epoch 54: Train Loss: 0.0128, Train Acc: 1.0000, Val Loss: 0.0401, Val Acc: 1.0000

Epoch 55: Train Loss: 0.0107, Train Acc: 1.0000, Val Loss: 0.0371, Val Acc: 1.0000

Epoch 56: Train Loss: 0.0098, Train Acc: 1.0000, Val Loss: 0.0374, Val Acc: 1.0000

Epoch 57: Train Loss: 0.0096, Train Acc: 1.0000, Val Loss: 0.0357, Val Acc: 1.0000

Epoch 58: Train Loss: 0.0092, Train Acc: 1.0000, Val Loss: 0.0313, Val Acc: 1.0000

Epoch 59: Train Loss: 0.0075, Train Acc: 1.0000, Val Loss: 0.0304, Val Acc: 1.0000

Epoch 60: Train Loss: 0.0065, Train Acc: 1.0000, Val Loss: 0.0253, Val Acc: 1.0000

Epoch 61: Train Loss: 0.0073, Train Acc: 1.0000, Val Loss: 0.0265, Val Acc: 1.0000

Epoch 62: Train Loss: 0.0070, Train Acc: 1.0000, Val Loss: 0.0242, Val Acc: 1.0000

Epoch 63: Train Loss: 0.0063, Train Acc: 1.0000, Val Loss: 0.0297, Val Acc: 1.0000

Epoch 64: Train Loss: 0.0057, Train Acc: 1.0000, Val Loss: 0.0236, Val Acc: 1.0000

Epoch 65: Train Loss: 0.0059, Train Acc: 1.0000, Val Loss: 0.0219, Val Acc: 1.0000

Epoch 66: Train Loss: 0.0056, Train Acc: 1.0000, Val Loss: 0.0218, Val Acc: 1.0000

Epoch 67: Train Loss: 0.0050, Train Acc: 1.0000, Val Loss: 0.0232, Val Acc: 1.0000

Epoch 68: Train Loss: 0.0049, Train Acc: 1.0000, Val Loss: 0.0220, Val Acc: 1.0000

Epoch 69: Train Loss: 0.0048, Train Acc: 1.0000, Val Loss: 0.0186, Val Acc: 1.0000

Epoch 70: Train Loss: 0.0043, Train Acc: 1.0000, Val Loss: 0.0198, Val Acc: 1.0000

Epoch 71: Train Loss: 0.0038, Train Acc: 1.0000, Val Loss: 0.0196, Val Acc: 1.0000

Epoch 72: Train Loss: 0.0045, Train Acc: 1.0000, Val Loss: 0.0186, Val Acc: 1.0000

Epoch 73: Train Loss: 0.0040, Train Acc: 1.0000, Val Loss: 0.0187, Val Acc: 1.0000

Epoch 74: Train Loss: 0.0035, Train Acc: 1.0000, Val Loss: 0.0187, Val Acc:

```

1.0000
Epoch 75: Train Loss: 0.0041, Train Acc: 1.0000, Val Loss: 0.0185, Val Acc:
1.0000
Epoch 76: Train Loss: 0.0035, Train Acc: 1.0000, Val Loss: 0.0179, Val Acc:
1.0000
Epoch 77: Train Loss: 0.0032, Train Acc: 1.0000, Val Loss: 0.0172, Val Acc:
1.0000
Epoch 78: Train Loss: 0.0029, Train Acc: 1.0000, Val Loss: 0.0174, Val Acc:
1.0000
Epoch 79: Train Loss: 0.0035, Train Acc: 1.0000, Val Loss: 0.0165, Val Acc:
1.0000
Epoch 80: Train Loss: 0.0029, Train Acc: 1.0000, Val Loss: 0.0168, Val Acc:
1.0000
Epoch 81: Train Loss: 0.0026, Train Acc: 1.0000, Val Loss: 0.0171, Val Acc:
1.0000
Epoch 82: Train Loss: 0.0032, Train Acc: 1.0000, Val Loss: 0.0165, Val Acc:
1.0000
Epoch 83: Train Loss: 0.0026, Train Acc: 1.0000, Val Loss: 0.0163, Val Acc:
1.0000
Epoch 84: Train Loss: 0.0027, Train Acc: 1.0000, Val Loss: 0.0155, Val Acc:
1.0000
Epoch 85: Train Loss: 0.0023, Train Acc: 1.0000, Val Loss: 0.0159, Val Acc:
1.0000
Epoch 86: Train Loss: 0.0029, Train Acc: 1.0000, Val Loss: 0.0153, Val Acc:
1.0000
Epoch 87: Train Loss: 0.0023, Train Acc: 1.0000, Val Loss: 0.0154, Val Acc:
1.0000
Epoch 88: Train Loss: 0.0024, Train Acc: 1.0000, Val Loss: 0.0145, Val Acc:
1.0000
Epoch 89: Train Loss: 0.0021, Train Acc: 1.0000, Val Loss: 0.0149, Val Acc:
1.0000
Epoch 90: Train Loss: 0.0025, Train Acc: 1.0000, Val Loss: 0.0144, Val Acc:
1.0000
Epoch 91: Train Loss: 0.0021, Train Acc: 1.0000, Val Loss: 0.0146, Val Acc:
1.0000
Early stopping triggered

```

[27]: `summary_nn(model)`

```

MyModel(
  (network): Sequential(
    (0): Linear(in_features=2, out_features=16, bias=True)
    (1): ReLU()
    (2): Linear(in_features=16, out_features=16, bias=True)
    (3): ReLU()
    (4): Linear(in_features=16, out_features=2, bias=True)
  )
)

```

```

Param: network.0.weight Shape: [16, 2] Numel: 32
Param: network.0.bias Shape: [16] Numel: 16
Param: network.2.weight Shape: [16, 16] Numel: 256
Param: network.2.bias Shape: [16] Numel: 16
Param: network.4.weight Shape: [2, 16] Numel: 32
Param: network.4.bias Shape: [2] Numel: 2
Total Numel = 354

```

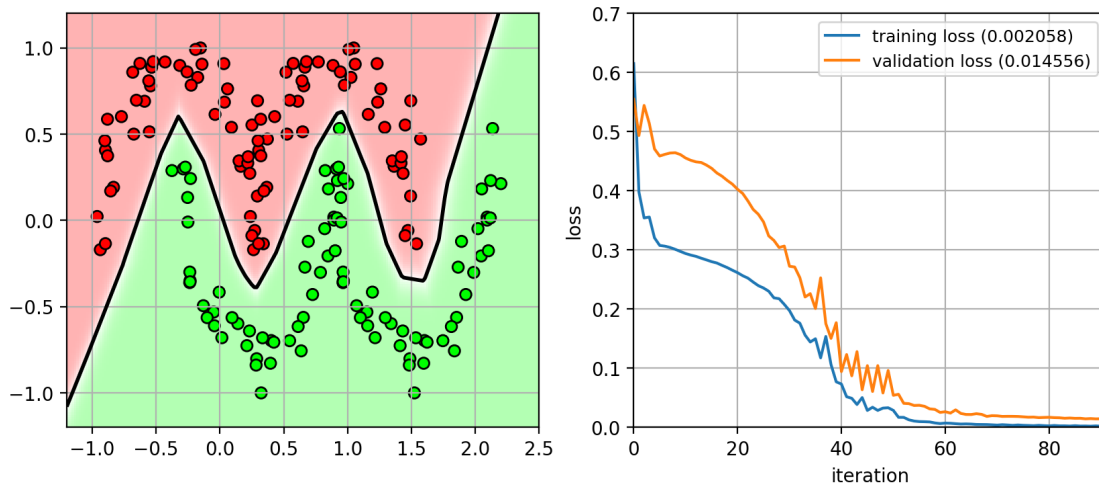
```

[28]: nnfig = plt.figure(figsize=(10,4))
      plt.subplot(1,2,1)
      plot_nn(model, axbox2, trainX2, trainY2)
      plt.subplot(1,2,2)
      plot_history(history)
      plt.close()

nnfig

```

[28]:



18 We should use deeper networks...

- Less parameters than a 1 hidden-layer NN
 - but the number of parameters is still large
- Dataset is still too small.
- *Vanishing Gradient problem*
 - backprop recursively multiplies gradients
 - * numerical values get smaller.
 - * gradient signal is “washed” out the further back it travels.
- We’ll see how to address these problems later.

19 Example on MNIST Dataset

- Images are 28x28, digits 0-9
 - 6,000 for training
 - 10,000 for testing

```
[29]: def show_imgs(W_list, nc=10, highlight_green=None, highlight_red=None, titles=None):
    nfilter = len(W_list)
    nr = (nfilter - 1) // nc + 1
    for i in range(nr):
        for j in range(nc):
            idx = i * nc + j
            if idx == nfilter:
                break
            plt.subplot(nr, nc, idx + 1)
            cur_W = W_list[idx]
            plt.imshow(cur_W, cmap='gray', interpolation='nearest')
            if titles is not None:
                plt.title(titles % idx)

            if ((highlight_green is not None) and highlight_green[idx]) or \
                ((highlight_red is not None) and highlight_red[idx]):
                ax = plt.gca()
                if highlight_green[idx]:
                    mycol = '#00FF00'
                else:
                    mycol = 'r'
                for S in ['bottom', 'top', 'right', 'left']:
                    ax.spines[S].set_color(mycol)
                    ax.spines[S].set_lw(2.0)
                ax.xaxis.set_ticks_position('none')
                ax.yaxis.set_ticks_position('none')
                ax.set_xticks([])
                ax.set_yticks([])
            else:
                plt.gca().set_axis_off()
```

```
[30]: import struct

def read_32int(f):
    return struct.unpack('>I', f.read(4))[0]
def read_img(img_path):
    with open(img_path, 'rb') as f:
        magic_num = read_32int(f)
        num_image = read_32int(f)
        n_row = read_32int(f)
        n_col = read_32int(f)
```



```

        #print 'num_image = {}; n_row = {}; n_col = {}'.format(num_image,
↪n_row, n_col)
        res = []
        npixel = n_row * n_col
        res_arr = fromfile(f, dtype='B')
        res_arr = res_arr.reshape((num_image, n_row, n_col), order='C')
        #print 'image data shape = {}'.format(res_arr.shape)
        return num_image, n_row, n_col, res_arr
def read_label(label_path):
    with open(label_path, 'rb') as f:
        magic_num = read_32int(f)
        num_label = read_32int(f)
        #print 'num_label = {}'.format(num_label)
        res_arr = fromfile(f, dtype='B')
        #print res_arr.shape
        #res_arr = res_arr.reshape((num_label, 1))
        res_arr = res_arr.ravel()
        #print 'label data shape = {}'.format(res_arr.shape)
        return num_label, res_arr

```

```

[31]: n_train, nrow, ncol, training = read_img('data/train-images.idx3-ubyte')
_, trainY = read_label('data/train-labels.idx1-ubyte')
n_test, _, _, testing = read_img('data/t10k-images.idx3-ubyte')
_, testY = read_label('data/t10k-labels.idx1-ubyte')

# for demonstration we only use 10% of the training data
sample_index = range(0, training.shape[0], 10)
training = training[sample_index]
trainY = trainY[sample_index]
print(training.shape)
print(trainY.shape)
print(testing.shape)
print(testY.shape)

```

```

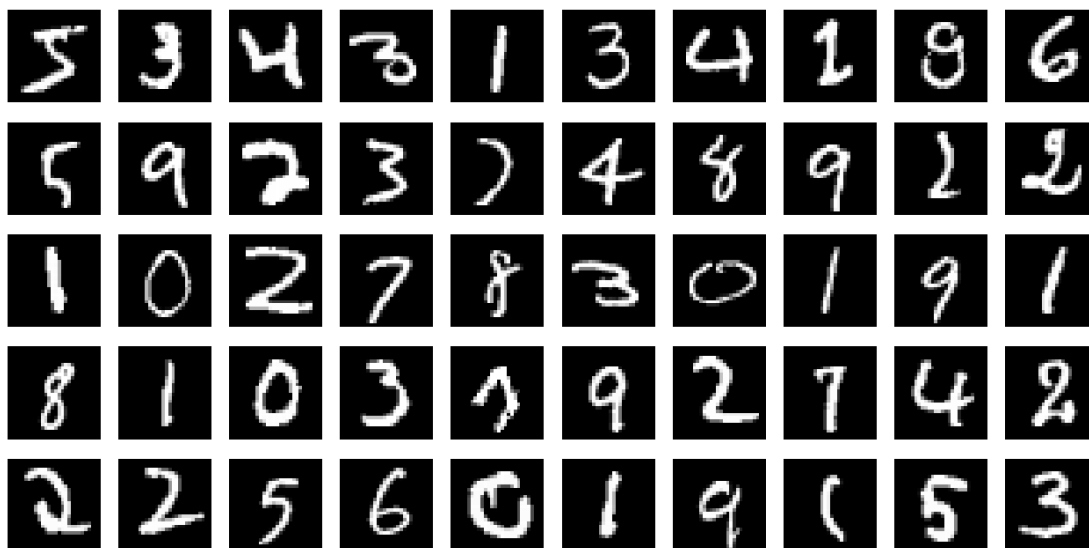
(6000, 28, 28)
(6000,)
(10000, 28, 28)
(10000,)

```

```

[32]: # Example images
plt.figure(figsize=(8,4))
show_imgs(training[0:50])

```



20 Pre-processing

- Reshape images into vectors
- map to $[0,1]$, then subtract the mean

```
[33]: # Reshape the images to a vector
# and map the data to [0,1]
trainXraw = training.reshape((len(training), -1), order='C') / 255.0
testXraw = testing.reshape((len(testing), -1), order='C') / 255.0

# center the image data (but don't change variance)
scaler = preprocessing.StandardScaler(with_std=False)
trainX = scaler.fit_transform(trainXraw).astype(np.float32)
testX = scaler.transform(testXraw).astype(np.float32)

print(trainX.shape)
print(trainY.shape)
```

(6000, 784)

(6000,)

- Generate a fixed validation set
 - use vtrainX for training and validX for validation

```
[34]: # generate a fixed validation set using 10% of the training set
vtrainX, validX, vtrainY, validY = \
    model_selection.train_test_split(trainX, trainY,
                                     train_size=0.9, test_size=0.1, random_state=4487)
```

```
# validation data
validset = (validX, validY)
```

21 MNIST - Logistic Regression (0-hidden layers)

- Training procedure
 - We specify the validation set so that it will be fixed when we change the `random_state` to randomly initialize the weights.
 - Train on the non-validation training data.
 - Use a larger batch size to speed up the algorithm

```
[35]: def plot_history2(history):
    fig, ax1 = plt.subplots()
    ax1.plot(history['train_loss'], 'r', label="training loss ({:.6f})".
    ↪format(history['train_loss'][-1]))
    ax1.plot(history['val_loss'], 'r--', label="validation loss ({:.6f})".
    ↪format(history['val_loss'][-1]))
    ax1.grid(True)
    ax1.set_xlabel('iteration')
    ax1.legend(loc="best", fontsize=9)
    ax1.set_ylabel('loss', color='r')
    ax1.tick_params('y', colors='r')
    if 'train_acc' in history:
        ax2 = ax1.twinx()
        ax2.plot(history['train_acc'], 'b', label="training acc ({:.4f})".
        ↪format(history['train_acc'][-1]))
        ax2.plot(history['val_acc'], 'b--', label="validation acc ({:.4f})".
        ↪format(history['val_acc'][-1]))
        ax2.legend(loc="best", fontsize=9)
        ax2.set_ylabel('acc', color='b')
        ax2.tick_params('y', colors='b')
```

```
[36]: model = MyModel(input_size=784, output_size=10, hidden_sizes=[])
criterion = nn.CrossEntropyLoss() # loss function
optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True) ↪
    ↪# optimizer
train_loader = get_dataloader(vtrainX, vtrainY) # data loaders
val_loader = get_dataloader(validX, validY)
# train the model
history = train_model(model, criterion, train_loader, val_loader, ↪
    ↪total_epoch=100, patience=10)
```

Epoch 1: Train Loss: 0.5367, Train Acc: 0.8357, Val Loss: 0.3920, Val Acc: 0.8867

Epoch 2: Train Loss: 0.3476, Train Acc: 0.9017, Val Loss: 0.3694, Val Acc: 0.8950

Epoch 3: Train Loss: 0.3062, Train Acc: 0.9130, Val Loss: 0.3644, Val Acc:

```

0.8967
Epoch 4: Train Loss: 0.2801, Train Acc: 0.9207, Val Loss: 0.3647, Val Acc:
0.8950
Epoch 5: Train Loss: 0.2609, Train Acc: 0.9283, Val Loss: 0.3670, Val Acc:
0.8900
Epoch 6: Train Loss: 0.2456, Train Acc: 0.9344, Val Loss: 0.3702, Val Acc:
0.8850
Epoch 7: Train Loss: 0.2328, Train Acc: 0.9372, Val Loss: 0.3738, Val Acc:
0.8833
Epoch 8: Train Loss: 0.2219, Train Acc: 0.9407, Val Loss: 0.3776, Val Acc:
0.8833
Epoch 9: Train Loss: 0.2123, Train Acc: 0.9422, Val Loss: 0.3816, Val Acc:
0.8850
Epoch 10: Train Loss: 0.2038, Train Acc: 0.9439, Val Loss: 0.3856, Val Acc:
0.8850
Epoch 11: Train Loss: 0.1960, Train Acc: 0.9448, Val Loss: 0.3897, Val Acc:
0.8850
Epoch 12: Train Loss: 0.1890, Train Acc: 0.9474, Val Loss: 0.3938, Val Acc:
0.8867
Epoch 13: Train Loss: 0.1826, Train Acc: 0.9496, Val Loss: 0.3979, Val Acc:
0.8867
Early stopping triggered

```

```

[37]: summary_nn(model)
      plot_history2(history)

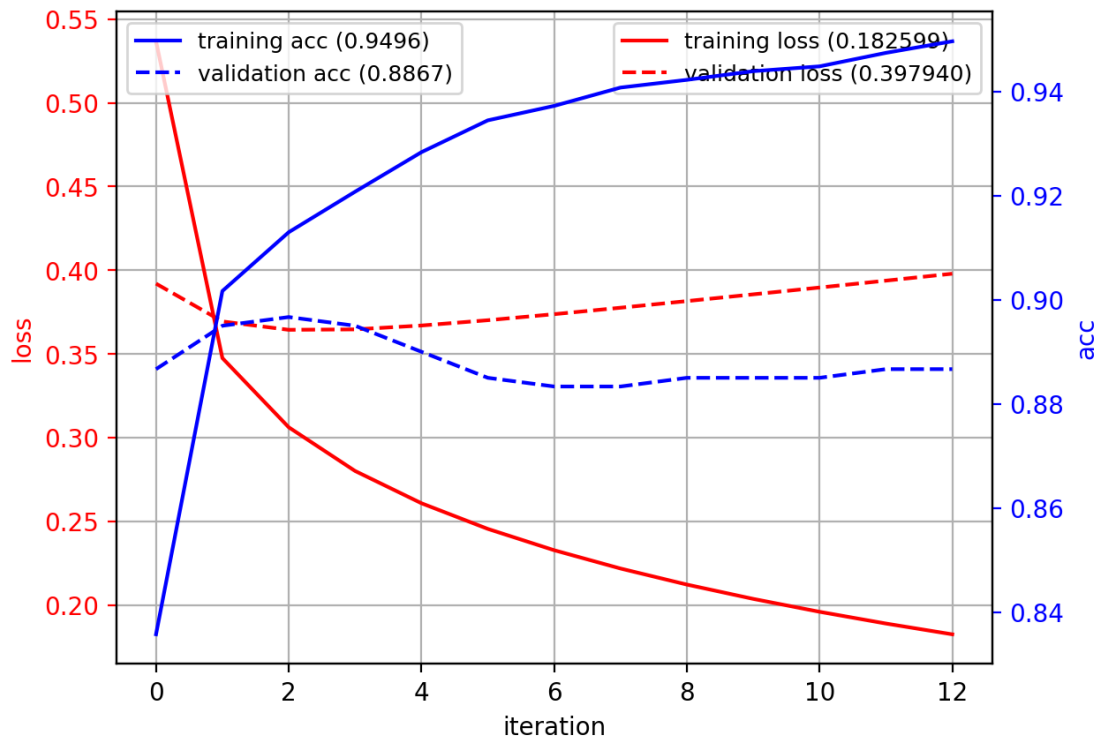
      with torch.no_grad():
          predY = model(torch.as_tensor(testX)).argmax(-1)
      acc = metrics.accuracy_score(testY, predY)
      print("test accuracy:", acc)

```

```

MyModel(
  (network): Sequential(
    (0): Linear(in_features=784, out_features=10, bias=True)
  )
)
Param: network.0.weight Shape: [10, 784]      Numel: 7840
Param: network.0.bias   Shape: [10]           Numel: 10
Total Numel = 7850
test accuracy: 0.8813

```

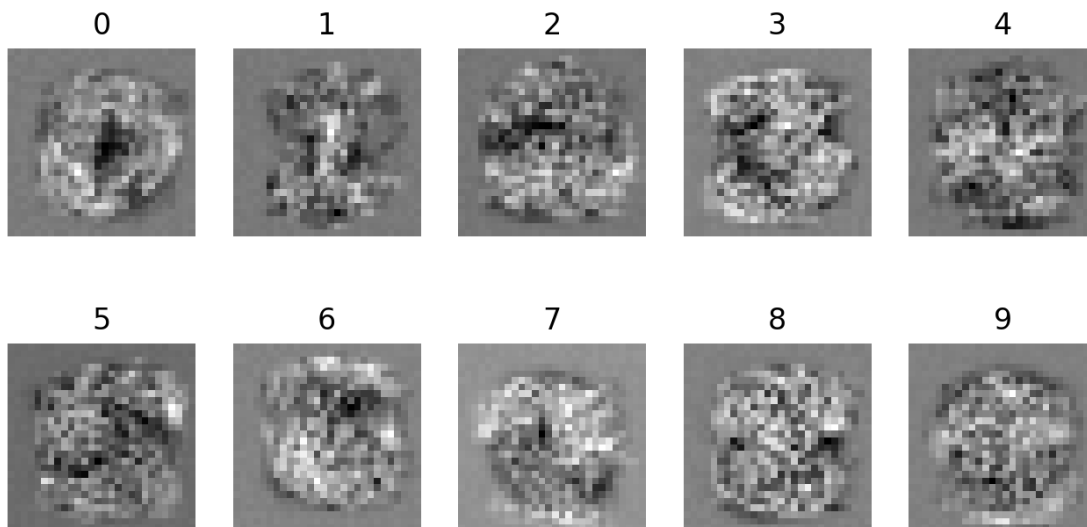


```
[38]: params = model.network[0].weight.data
      print(params)
```

```
tensor([[ 3.3546e-02, -3.4324e-02,  2.0939e-02, ..., -2.0905e-02,
          1.1012e-02, -3.1561e-02],
        [ 2.7129e-02,  3.1136e-03, -1.7196e-02, ..., -1.5344e-02,
          -2.0055e-02, -2.7852e-03],
        [ 2.9864e-02, -3.4973e-02, -1.7804e-02, ...,  1.1404e-02,
          -8.0777e-03, -5.1047e-05],
        ...,
        [-1.3821e-02, -1.6467e-03, -2.6531e-03, ...,  2.2454e-02,
          1.5731e-02,  5.1474e-03],
        [-4.0478e-03,  2.0415e-02, -5.3243e-03, ...,  2.2182e-02,
          -2.7403e-02, -2.7426e-02],
        [-2.7465e-02, -8.7818e-03,  2.0570e-02, ..., -5.8698e-05,
          -1.5991e-02, -1.2174e-04]])
```

- Reshape the weights into an image
 - input images that match the weights will have high response for that class.

```
[39]: filter_list = [w.view((28,28)) for w in params]
      plt.figure(figsize=(8,4))
      show_imgs(filter_list, nc=5, titles="%d")
```



22 MNIST - 1-hidden layer

- Add 1 hidden layer with 50 ReLu nodes
 - each node is extracting a feature from the input image

```
[40]: model = MyModel(input_size=784, output_size=10, hidden_sizes=[50])
      criterion = nn.CrossEntropyLoss() # loss function
      optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True)
      ↪ # optimizer
      train_loader = get_dataloader(vtrainX, vtrainY) # data loaders
      val_loader = get_dataloader(validX, validY)
      # train the model
      history = train_model(model, criterion, train_loader, val_loader,
      ↪ total_epoch=100, patience=10)
      # summary and plot
      summary_nn(model)
      plot_history2(history)
      # test
      with torch.no_grad():
          predY = model(torch.as_tensor(testX)).argmax(-1)
      acc = metrics.accuracy_score(testY, predY)
      print("test accuracy:", acc)
```

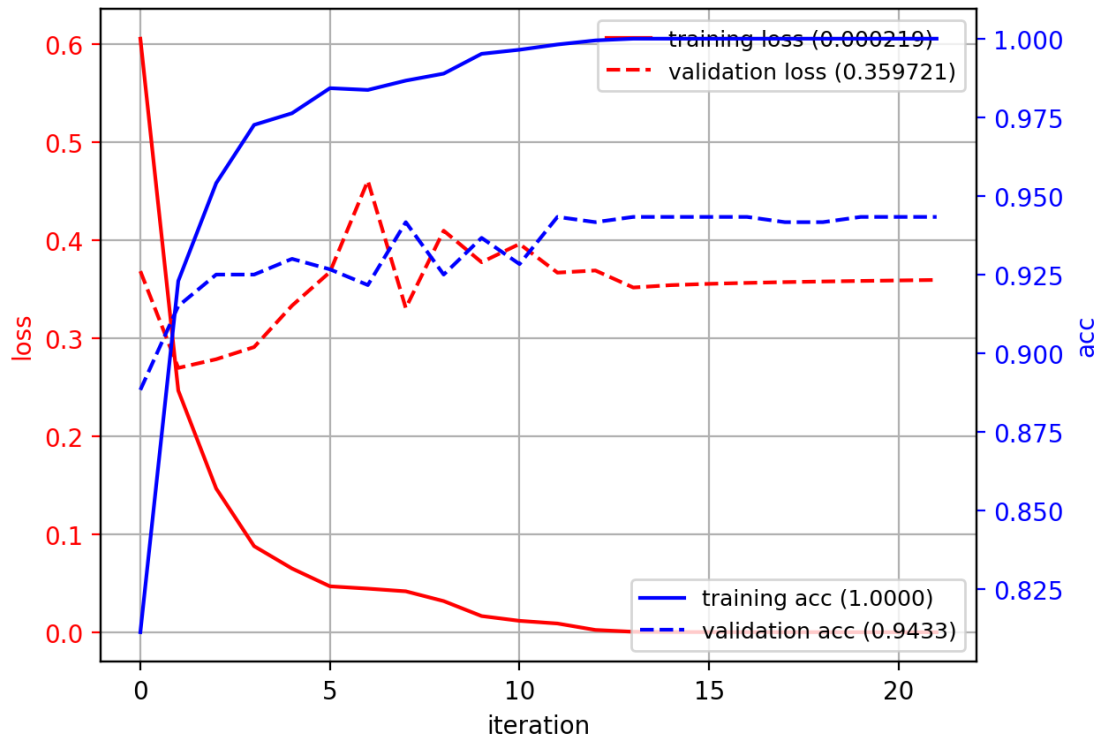
```
Epoch 1: Train Loss: 0.6059, Train Acc: 0.8113, Val Loss: 0.3687, Val Acc:
0.8883
Epoch 2: Train Loss: 0.2468, Train Acc: 0.9230, Val Loss: 0.2699, Val Acc:
0.9150
Epoch 3: Train Loss: 0.1468, Train Acc: 0.9541, Val Loss: 0.2788, Val Acc:
0.9250
```

```

Epoch 4: Train Loss: 0.0880, Train Acc: 0.9726, Val Loss: 0.2912, Val Acc:
0.9250
Epoch 5: Train Loss: 0.0652, Train Acc: 0.9763, Val Loss: 0.3335, Val Acc:
0.9300
Epoch 6: Train Loss: 0.0471, Train Acc: 0.9843, Val Loss: 0.3678, Val Acc:
0.9267
Epoch 7: Train Loss: 0.0447, Train Acc: 0.9837, Val Loss: 0.4609, Val Acc:
0.9217
Epoch 8: Train Loss: 0.0420, Train Acc: 0.9867, Val Loss: 0.3310, Val Acc:
0.9417
Epoch 9: Train Loss: 0.0320, Train Acc: 0.9889, Val Loss: 0.4100, Val Acc:
0.9250
Epoch 10: Train Loss: 0.0167, Train Acc: 0.9952, Val Loss: 0.3778, Val Acc:
0.9367
Epoch 11: Train Loss: 0.0118, Train Acc: 0.9965, Val Loss: 0.3964, Val Acc:
0.9283
Epoch 12: Train Loss: 0.0091, Train Acc: 0.9981, Val Loss: 0.3672, Val Acc:
0.9433
Epoch 13: Train Loss: 0.0025, Train Acc: 0.9994, Val Loss: 0.3694, Val Acc:
0.9417
Epoch 14: Train Loss: 0.0007, Train Acc: 1.0000, Val Loss: 0.3520, Val Acc:
0.9433
Epoch 15: Train Loss: 0.0004, Train Acc: 1.0000, Val Loss: 0.3544, Val Acc:
0.9433
Epoch 16: Train Loss: 0.0004, Train Acc: 1.0000, Val Loss: 0.3558, Val Acc:
0.9433
Epoch 17: Train Loss: 0.0003, Train Acc: 1.0000, Val Loss: 0.3567, Val Acc:
0.9433
Epoch 18: Train Loss: 0.0003, Train Acc: 1.0000, Val Loss: 0.3575, Val Acc:
0.9417
Epoch 19: Train Loss: 0.0003, Train Acc: 1.0000, Val Loss: 0.3581, Val Acc:
0.9417
Epoch 20: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.3587, Val Acc:
0.9433
Epoch 21: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.3592, Val Acc:
0.9433
Epoch 22: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.3597, Val Acc:
0.9433
Early stopping triggered
MyModel(
  (network): Sequential(
    (0): Linear(in_features=784, out_features=50, bias=True)
    (1): ReLU()
    (2): Linear(in_features=50, out_features=10, bias=True)
  )
)
Param: network.0.weight Shape: [50, 784]          Numel: 39200
Param: network.0.bias   Shape: [50]              Numel: 50

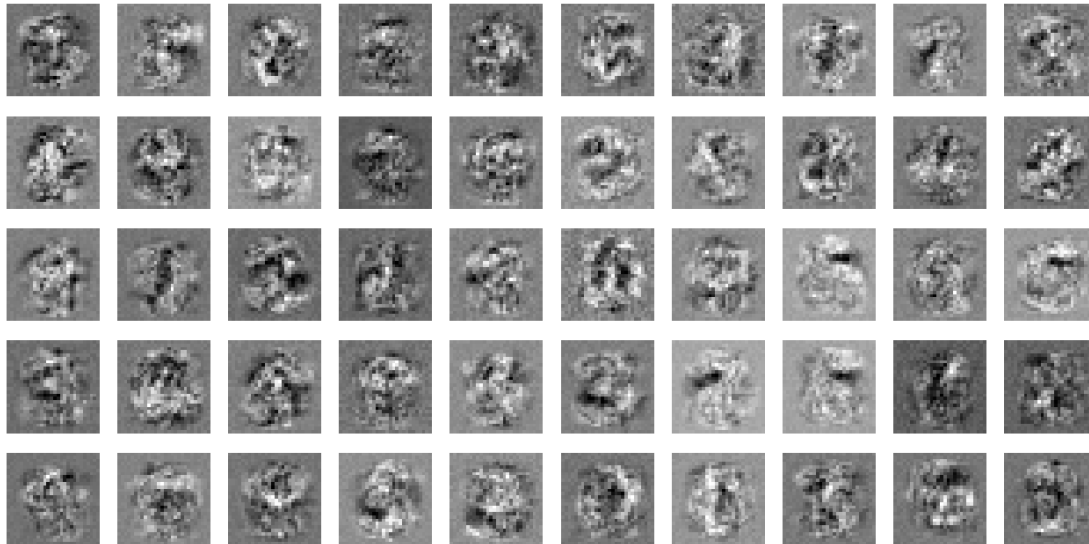
```

Param: network.2.weight Shape: [10, 50] Numel: 500
 Param: network.2.bias Shape: [10] Numel: 10
 Total Numel = 39760
 test accuracy: 0.9361



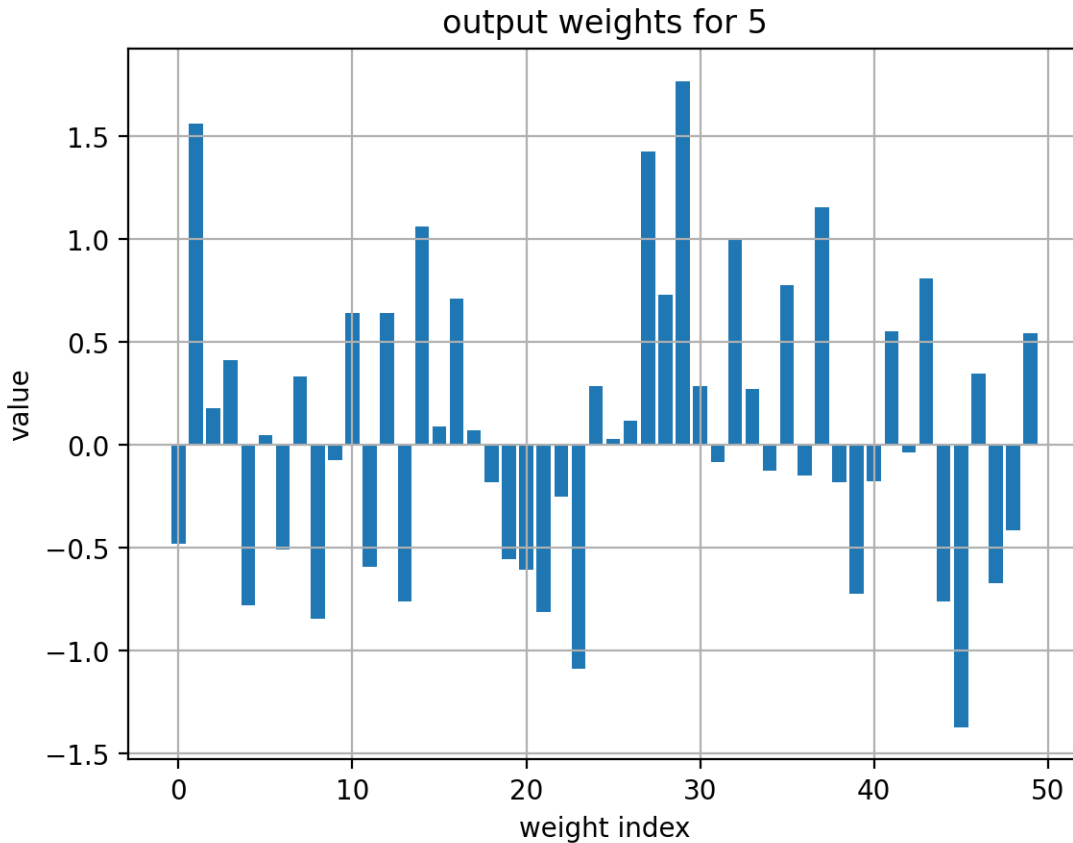
- Examine the weights of the hidden layer
 - $h_i = \sigma(\mathbf{w}_i^T \mathbf{x})$
 - each weight vector is a “pattern prototype” that the node will match
- The hidden nodes look for local structures:
 - oriented edges, curves, other local structures

```
[41]: W = model.network[0].weight.data
filter_list = [w.view((28,28)) for w in W]
plt.figure(figsize=(8,4))
show_imgs(filter_list, nc=10)
```

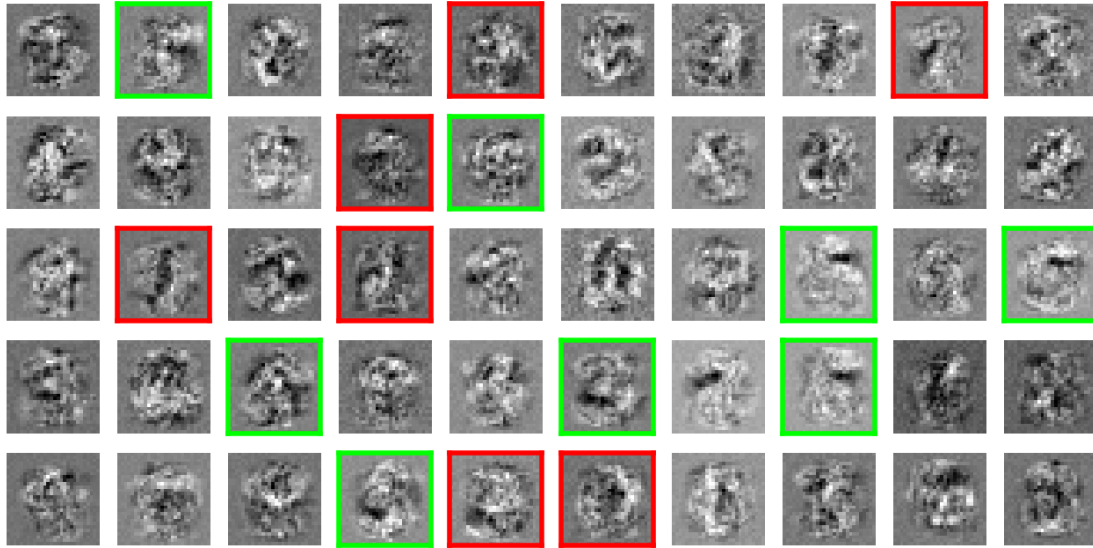
- Examine the weights of the 2nd layer (output)
 - $y_j = \sigma(\mathbf{w}_j^T \mathbf{h})$
 - recall the hidden-layer outputs h are always non-negative.
 - * positive value in $\mathbf{w}_j \rightarrow$ class j should have j -th pattern
 - * negative value in $\mathbf{w}_j \rightarrow$ class j shouldn't have j -th pattern

```
[42]: W = model.network[2].weight.data.T # model.network[1] is the relu
      # print(W.shape)
      d = 5
      plt.bar(arange(0,W.shape[0]),W[:,d]); plt.grid(True);
      plt.xlabel('weight index'); plt.ylabel('value')
      plt.title('output weights for {}'.format(d));
```



- For “5”, finds local image parts that correspond to 5
 - should have (green boxes):
 - * horizontal line at top; semicircle on the bottom
 - shouldn't have (red boxes):
 - * vertical line in top-right; verticle line in the middle

```
[43]: plt.figure(figsize=(8,4))
show_imgs(filter_list, nc=10,
          highlight_green=(W[:,d]>0.75), # positive weights are green
          highlight_red=(W[:,d]<-0.75)) # negative weights are red
```



23 1 Hidden layer with more nodes

- hidden layer with 200 nodes

```
[44]: model = MyModel(input_size=784, output_size=10, hidden_sizes=[200])
      criterion = nn.CrossEntropyLoss() # loss function
      optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True)
      ↪# optimizer
      train_loader = get_dataloader(vtrainX, vtrainY) # data loaders
      val_loader = get_dataloader(validX, validY)
      # train the model
      history = train_model(model, criterion, train_loader, val_loader,
      ↪total_epoch=100, patience=10)
      # summary and plot
      summary_nn(model)
      plot_history2(history)
      # test
      with torch.no_grad():
          predY = model(torch.as_tensor(testX)).argmax(-1)
      acc = metrics.accuracy_score(testY, predY)
      print("test accuracy:", acc)
```

Epoch 1: Train Loss: 0.5805, Train Acc: 0.8207, Val Loss: 0.3409, Val Acc: 0.8933

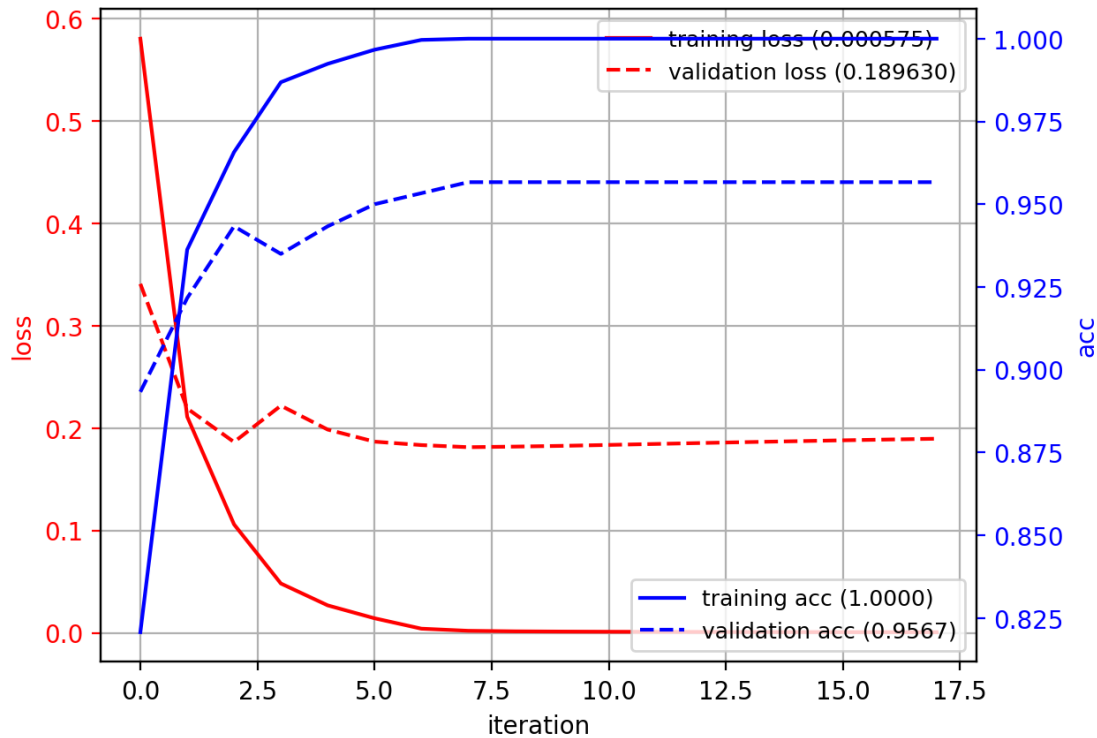
Epoch 2: Train Loss: 0.2112, Train Acc: 0.9363, Val Loss: 0.2192, Val Acc: 0.9217

Epoch 3: Train Loss: 0.1060, Train Acc: 0.9657, Val Loss: 0.1864, Val Acc: 0.9433

```

Epoch 4: Train Loss: 0.0482, Train Acc: 0.9869, Val Loss: 0.2219, Val Acc:
0.9350
Epoch 5: Train Loss: 0.0268, Train Acc: 0.9924, Val Loss: 0.1986, Val Acc:
0.9433
Epoch 6: Train Loss: 0.0142, Train Acc: 0.9967, Val Loss: 0.1868, Val Acc:
0.9500
Epoch 7: Train Loss: 0.0040, Train Acc: 0.9996, Val Loss: 0.1834, Val Acc:
0.9533
Epoch 8: Train Loss: 0.0020, Train Acc: 1.0000, Val Loss: 0.1814, Val Acc:
0.9567
Epoch 9: Train Loss: 0.0014, Train Acc: 1.0000, Val Loss: 0.1819, Val Acc:
0.9567
Epoch 10: Train Loss: 0.0012, Train Acc: 1.0000, Val Loss: 0.1827, Val Acc:
0.9567
Epoch 11: Train Loss: 0.0010, Train Acc: 1.0000, Val Loss: 0.1836, Val Acc:
0.9567
Epoch 12: Train Loss: 0.0009, Train Acc: 1.0000, Val Loss: 0.1845, Val Acc:
0.9567
Epoch 13: Train Loss: 0.0008, Train Acc: 1.0000, Val Loss: 0.1855, Val Acc:
0.9567
Epoch 14: Train Loss: 0.0008, Train Acc: 1.0000, Val Loss: 0.1864, Val Acc:
0.9567
Epoch 15: Train Loss: 0.0007, Train Acc: 1.0000, Val Loss: 0.1872, Val Acc:
0.9567
Epoch 16: Train Loss: 0.0007, Train Acc: 1.0000, Val Loss: 0.1881, Val Acc:
0.9567
Epoch 17: Train Loss: 0.0006, Train Acc: 1.0000, Val Loss: 0.1889, Val Acc:
0.9567
Epoch 18: Train Loss: 0.0006, Train Acc: 1.0000, Val Loss: 0.1896, Val Acc:
0.9567
Early stopping triggered
MyModel(
  (network): Sequential(
    (0): Linear(in_features=784, out_features=200, bias=True)
    (1): ReLU()
    (2): Linear(in_features=200, out_features=10, bias=True)
  )
)
Param: network.0.weight Shape: [200, 784]      Numel: 156800
Param: network.0.bias   Shape: [200]          Numel: 200
Param: network.2.weight Shape: [10, 200]       Numel: 2000
Param: network.2.bias   Shape: [10]           Numel: 10
Total Numel = 159010
test accuracy: 0.945

```



- hidden layer with 1000 nodes

```
[45]: model = MyModel(input_size=784, output_size=10, hidden_sizes=[1000])
criterion = nn.CrossEntropyLoss() # loss function
optimizer = optim.SGD(model.parameters(), lr=0.1, momentum=0.9, nesterov=True)
    ↪ # optimizer
train_loader = get_dataloader(vtrainX, vtrainY) # data loaders
val_loader = get_dataloader(validX, validY)
# train the model
history = train_model(model, criterion, train_loader, val_loader,
    ↪ total_epoch=100, patience=10)
# summary and plot
summary_nn(model)
plot_history2(history)
# test
with torch.no_grad():
    predY = model(torch.as_tensor(testX)).argmax(-1)
acc = metrics.accuracy_score(testY, predY)
print("test accuracy:", acc)
```

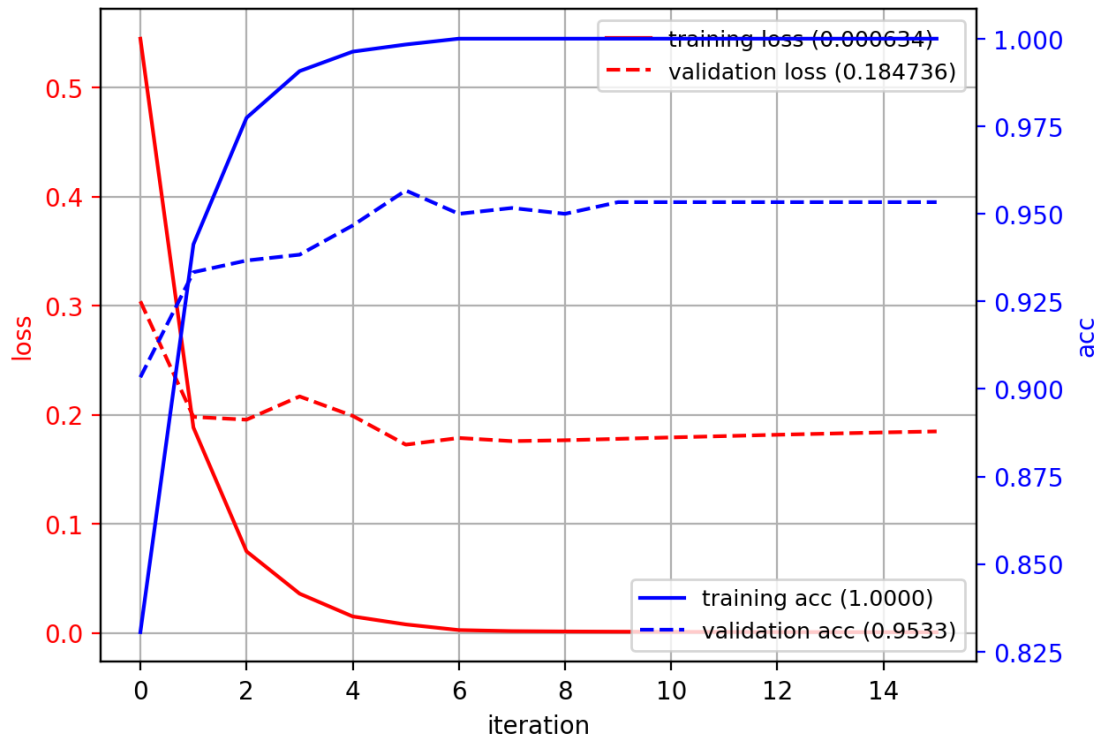
Epoch 1: Train Loss: 0.5448, Train Acc: 0.8306, Val Loss: 0.3040, Val Acc: 0.9033

Epoch 2: Train Loss: 0.1883, Train Acc: 0.9413, Val Loss: 0.1980, Val Acc: 0.9333

```

Epoch 3: Train Loss: 0.0748, Train Acc: 0.9774, Val Loss: 0.1955, Val Acc:
0.9367
Epoch 4: Train Loss: 0.0360, Train Acc: 0.9907, Val Loss: 0.2168, Val Acc:
0.9383
Epoch 5: Train Loss: 0.0151, Train Acc: 0.9963, Val Loss: 0.1990, Val Acc:
0.9467
Epoch 6: Train Loss: 0.0078, Train Acc: 0.9983, Val Loss: 0.1726, Val Acc:
0.9567
Epoch 7: Train Loss: 0.0026, Train Acc: 1.0000, Val Loss: 0.1787, Val Acc:
0.9500
Epoch 8: Train Loss: 0.0016, Train Acc: 1.0000, Val Loss: 0.1758, Val Acc:
0.9517
Epoch 9: Train Loss: 0.0013, Train Acc: 1.0000, Val Loss: 0.1766, Val Acc:
0.9500
Epoch 10: Train Loss: 0.0011, Train Acc: 1.0000, Val Loss: 0.1779, Val Acc:
0.9533
Epoch 11: Train Loss: 0.0010, Train Acc: 1.0000, Val Loss: 0.1792, Val Acc:
0.9533
Epoch 12: Train Loss: 0.0009, Train Acc: 1.0000, Val Loss: 0.1804, Val Acc:
0.9533
Epoch 13: Train Loss: 0.0008, Train Acc: 1.0000, Val Loss: 0.1817, Val Acc:
0.9533
Epoch 14: Train Loss: 0.0007, Train Acc: 1.0000, Val Loss: 0.1828, Val Acc:
0.9533
Epoch 15: Train Loss: 0.0007, Train Acc: 1.0000, Val Loss: 0.1838, Val Acc:
0.9533
Epoch 16: Train Loss: 0.0006, Train Acc: 1.0000, Val Loss: 0.1847, Val Acc:
0.9533
Early stopping triggered
MyModel(
  (network): Sequential(
    (0): Linear(in_features=784, out_features=1000, bias=True)
    (1): ReLU()
    (2): Linear(in_features=1000, out_features=10, bias=True)
  )
)
Param: network.0.weight Shape: [1000, 784]      Numel: 784000
Param: network.0.bias   Shape: [1000]          Numel: 1000
Param: network.2.weight Shape: [10, 1000]       Numel: 10000
Param: network.2.bias   Shape: [10]             Numel: 10
Total Numel = 795010
test accuracy: 0.9487

```



- 2 hidden layers
 - input (28x28) -> 500 nodes -> 500 nodes -> output
 - Slightly better

```
[46]: model = MyModel(input_size=784, output_size=10, hidden_sizes=[500, 500])
      criterion = nn.CrossEntropyLoss() # loss function
      optimizer = optim.SGD(model.parameters(), lr=0.05, momentum=0.9, nesterov=True)
      ↪ # note: need to tune the learning rate!
      train_loader = get_dataloader(vtrainX, vtrainY) # data loaders
      val_loader = get_dataloader(validX, validY)
      # train the model
      history = train_model(model, criterion, train_loader, val_loader,
      ↪total_epoch=100, patience=10)
      # summary and plot
      summary_nn(model)
      plot_history2(history)
      # test
      with torch.no_grad():
          predY = model(torch.as_tensor(testX)).argmax(-1)
      acc = metrics.accuracy_score(testY, predY)
      print("test accuracy:", acc)
```

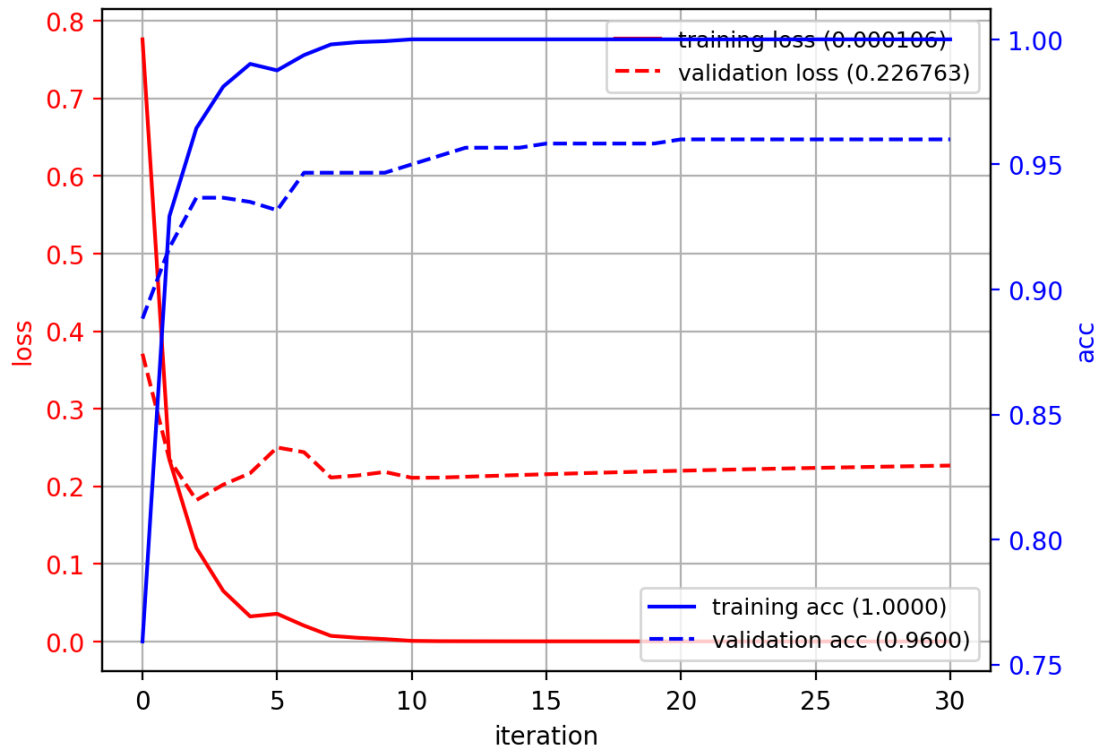
Epoch 1: Train Loss: 0.7761, Train Acc: 0.7593, Val Loss: 0.3708, Val Acc: 0.8883

Epoch 2: Train Loss: 0.2343, Train Acc: 0.9293, Val Loss: 0.2346, Val Acc: 0.9167
Epoch 3: Train Loss: 0.1207, Train Acc: 0.9644, Val Loss: 0.1822, Val Acc: 0.9367
Epoch 4: Train Loss: 0.0652, Train Acc: 0.9811, Val Loss: 0.2022, Val Acc: 0.9367
Epoch 5: Train Loss: 0.0323, Train Acc: 0.9902, Val Loss: 0.2170, Val Acc: 0.9350
Epoch 6: Train Loss: 0.0358, Train Acc: 0.9876, Val Loss: 0.2501, Val Acc: 0.9317
Epoch 7: Train Loss: 0.0205, Train Acc: 0.9937, Val Loss: 0.2440, Val Acc: 0.9467
Epoch 8: Train Loss: 0.0073, Train Acc: 0.9980, Val Loss: 0.2113, Val Acc: 0.9467
Epoch 9: Train Loss: 0.0048, Train Acc: 0.9989, Val Loss: 0.2141, Val Acc: 0.9467
Epoch 10: Train Loss: 0.0031, Train Acc: 0.9993, Val Loss: 0.2185, Val Acc: 0.9467
Epoch 11: Train Loss: 0.0008, Train Acc: 1.0000, Val Loss: 0.2110, Val Acc: 0.9500
Epoch 12: Train Loss: 0.0004, Train Acc: 1.0000, Val Loss: 0.2112, Val Acc: 0.9533
Epoch 13: Train Loss: 0.0004, Train Acc: 1.0000, Val Loss: 0.2123, Val Acc: 0.9567
Epoch 14: Train Loss: 0.0003, Train Acc: 1.0000, Val Loss: 0.2135, Val Acc: 0.9567
Epoch 15: Train Loss: 0.0003, Train Acc: 1.0000, Val Loss: 0.2146, Val Acc: 0.9567
Epoch 16: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2156, Val Acc: 0.9583
Epoch 17: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2166, Val Acc: 0.9583
Epoch 18: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2175, Val Acc: 0.9583
Epoch 19: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2184, Val Acc: 0.9583
Epoch 20: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2193, Val Acc: 0.9583
Epoch 21: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2201, Val Acc: 0.9600
Epoch 22: Train Loss: 0.0002, Train Acc: 1.0000, Val Loss: 0.2209, Val Acc: 0.9600
Epoch 23: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2217, Val Acc: 0.9600
Epoch 24: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2224, Val Acc: 0.9600
Epoch 25: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2231, Val Acc: 0.9600


```

Epoch 26: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2238, Val Acc:
0.9600
Epoch 27: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2244, Val Acc:
0.9600
Epoch 28: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2250, Val Acc:
0.9600
Epoch 29: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2256, Val Acc:
0.9600
Epoch 30: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2262, Val Acc:
0.9600
Epoch 31: Train Loss: 0.0001, Train Acc: 1.0000, Val Loss: 0.2268, Val Acc:
0.9600
Early stopping triggered
MyModel(
  (network): Sequential(
    (0): Linear(in_features=784, out_features=500, bias=True)
    (1): ReLU()
    (2): Linear(in_features=500, out_features=500, bias=True)
    (3): ReLU()
    (4): Linear(in_features=500, out_features=10, bias=True)
  )
)
Param: network.0.weight Shape: [500, 784]      Numel: 392000
Param: network.0.bias   Shape: [500]          Numel: 500
Param: network.2.weight Shape: [500, 500]      Numel: 250000
Param: network.2.bias   Shape: [500]          Numel: 500
Param: network.4.weight Shape: [10, 500]       Numel: 5000
Param: network.4.bias   Shape: [10]           Numel: 10
Total Numel = 648010
test accuracy: 0.9454

```



24 Comparison on MNIST

- Performance is saturated.

Type

No.Layers

Architecture

No.Parameters

Test Accuracy

LR

1

output(10)

7,850

0.8813

MLP

2

ReLu(50), output(10)

39,760

0.9395

MLP

2

Relu(200), output(10)

159,010

0.9464
MLP
2
Relu(1000), output(10)
795,010
0.9482
MLP
3
ReLU(500), Relu(500), output(10)
648,010
0.9463

25 Summary

- **Different types of neural networks**
 - *Perceptron* - single node (similar to logistic regression)
 - *Multi-layer perceptron (MLP)* - collection of perceptrons in layers
 - * also called *fully-connected layers* or *dense layers*
- **Training**
 - optimize loss function using stochastic gradient descent
- **Advantages**
 - lots of parameters - large capacity to learn from large amounts of data
- **Disadvantages**
 - lots of parameters - easy to overfit data
 - * need to monitor the training process
 - sensitive to initialization, learning rate, training algorithm.

26 Other things

- **Numerical stability**
 - normalize the inputs to $[-1,1]$ or $[0,1]$
- **Improving speed**
 - parallelize computations using GPU (Nvidia+CUDA)
- **Initialization**
 - the resulting network is still sensitive to initialization.
 - * Solution: train several networks and combine them as an ensemble.
- **Training problems**
 - For very deep networks, the “vanishing gradient” problem can hinder convergence
 - lack of data
 - We will see how to address these problems later.

27 References

- Software
 - pytorch: <https://pytorch.org/>
 - see Canvas site for installation tips.

- History:
 - https://en.wikipedia.org/wiki/History_of_artificial_neural_networks
- Online courses:
 - <http://cs231n.github.io/neural-networks-1/>
 - <http://cs231n.github.io/convolutional-networks/>